TABLE OF CONTENTS

12 IMPACT ON CULTURAL HERITAGE 12-1

12.1 Introduction 12-1
12.2 Environmental Legislation, Standards and Guidelines 12-1
12.3 Assessment Scope and Methodology 12-2
12.4 Project Background 12-3
12.5 Built Heritage Impact Assessment 12-10
12.6 Archaeological Impact Assessment 12-33
12.7 References 12-50

LIST OF TABLES

Table 12.1 Built Heritage and Other Identified Items within the Assessment Area 12-12
Table 12.2 Summary of Potential Impact and the Proposed Mitigation Measures on Other Identified Items 12-24
Table 12.3 Summary of Potential Impact and the Proposed Mitigation Measures on Archaeological Heritage in the Vicinity of the Project Site 12-48

LIST OF FIGURES

Figure 12.1 Project Site Boundary and Assessment Area for Cultural Heritage Impact Assessment
Figure 12.2 Geological Map of Assessment Area
Figure 12.3 1963 Aerial Photo
Figure 12.4 1982 Aerial Photo
Figure 12.5 Previous Archaeological Works Conducted within Assessment Area (Key Plan)
Figure 12.5.1 Previous Archaeological Works Conducted within Assessment Area (Sheet 1 of 4)
Figure 12.5.2 Previous Archaeological Works Conducted within Assessment Area (Sheet 2 of 4)
Figure 12.5.3 Previous Archaeological Works Conducted within Assessment Area (Sheet 3 of 4)
Figure 12.5.4 Previous Archaeological Works Conducted within Assessment Area (Sheet 4 of 4)
Figure 12.6.1 Assessed Archaeological Potential Area Prior to This Study (1)
Figure 12.6.2 Assessed Archaeological Potential Area Prior to This Study (2)
Figure 12.6.3 Assessed Archaeological Potential Area Prior to This Study (3)
Figure 12.6.4 Assessed Archaeological Potential Area Prior to This Study (4)
Figure 12.7 Locations of Built Heritage and Other Identified Items (Key Plan)
Figure 12.7.1 Locations of Built Heritage and Other Identified Items (Sheet 1 of 12)
Figure 12.7.2 Locations of Built Heritage and Other Identified Items (Sheet 2 of 12)
Figure 12.7.3 Locations of Built Heritage and Other Identified Items (Sheet 3 of 12)
Figure 12.7.4 Locations of Built Heritage and Other Identified Items (Sheet 4 of 12)
Figure 12.7.5 Locations of Built Heritage and Other Identified Items (Sheet 5 of 12)
Figure 12.7.6 Locations of Built Heritage and Other Identified Items (Sheet 6 of 12)
Figure 12.7.7 Locations of Built Heritage and Other Identified Items (Sheet 7 of 12)
Figure 12.7.8 Locations of Built Heritage and Other Identified Items (Sheet 8 of 12)
Figure 12.7.9 Locations of Built Heritage and Other Identified Items (Sheet 9 of 12)
Figure 12.7.10 Locations of Built Heritage and Other Identified Items (Sheet 10 of 12)
Figure 12.7.11 Locations of Built Heritage and Other Identified Items (Sheet 11 of 12)
Figure 12.7.12 Locations of Built Heritage and Other Identified Items (Sheet 12 of 12)
Figure 12.8 Locations of Other Identified Items in Relations to RODP
Figure 12.9 Further Review of Past Disturbance that Affect Archaeological Potential Assessment
Figure 12.10 Overall Evaluated Archaeological Potential of Assessment Area
Figure 12.11 Mitigation Measures for Archaeological Heritage
Figure 12.12 Geology Map Overlay with Moderate-Low and High Archaeological Potential Areas within Project Site Requiring Mitigation Measures

LIST OF APPENDICES

Appendix 12.1 Built Heritage and Other Identified Items Recording Sheets
Appendix 12.2 Notable Visual References
Appendix 12.3 Site Visit Photos

12 IMPACT ON CULTURAL HERITAGE

12.1 Introduction

12.1.1 This section presents the findings of Cultural Heritage Impact Assessment (CHIA), covering a Built Heritage Impact Assessment (BHIA) and an Archaeological Impact Assessment (AIA). The potential direct and indirect impacts associated with the construction and operation of the Project on built heritage and archaeological heritage have been assessed in accordance with the requirements in Annexes 10 and 19 of the Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM), Section 3.4.12 and Appendix K of the EIA Study Brief (ESB-363/2023).

12.2 Environmental Legislation, Standards and Guidelines

12.2.1 Environmental legislation, standards and guidelines relevant to the CHIA include the following:

Related to EIA Study

- Environmental Impact Assessment Ordinance (EIAO) (Cap. 499);
- Annexes 10 and 19 of EIAO-TM; and

Others

- Antiquities and Monuments Ordinance (A&MO) (Cap. 53).

Environmental Impact Assessment Ordinance (EIAO) (Cap. 499) and EIAO-TM Annexes 10 and 19

12.2.2 According to Schedule 1 of the EIAO, “site of cultural heritage” means an antiquity or monument, whether being a place, building, site or structure or a relic, as defined in the A&MO (Cap. 53) and any place, building, site, or structure or a relic identified by the Antiquities and Monuments Office (AMO) to be of archaeological, historical or palaeontological significance.

12.2.3 Annexes 10 and 19 of EIAO-TM provide the criteria and guidelines for evaluating the impacts to sites of cultural heritage. It is stated in Annex 10 that all adverse impacts on sites of cultural heritage shall be kept to the absolute minimum and that the general presumption of impact assessment shall be in favour of the protection and conservation of all sites of cultural heritage. Annex 19 provides the commonly adopted approaches and methodologies for assessment of impact on sites of cultural heritage.

Antiquities and Monuments Ordinance (A&MO) (Cap. 53)

12.2.4 The Ordinance provides the statutory framework for preservation of objects of historical, archaeological and paleontological interest and for matters ancillary thereto or connected therewith. The Ordinance contains the statutory procedures for the Declaration of Monuments. Under the Ordinance, a “monument” means a place, building, site or structure which is declared to be a monument, historical building or archaeological or paleontological site or structure under Section 3 of the Ordinance. Excavations carried out on building works, demolition and interference of a proposed monument or monument are prohibited except under permit under Section 6 of the Ordinance.

12.3 Assessment Scope and Methodology

12.3.1 The CHIA, consisting of a BHIA and an AIA, has been conducted in accordance with the requirements given in Section 3.4.12 and Appendix K of the EIA Study Brief (No. ESB-363/2023), covering an assessment area of 300 m from the boundary of the Project Site (**Figure 12.1** refers). The BHIA is to assess the impacts on the known and unknown built heritage items within the assessment area, while the AIA is to evaluate the archaeological impact imposed by the Project and its associated works. The assessment methodology for BHIA and AIA are described below.

Built Heritage Impact Assessment

12.3.2 A desktop review was conducted to identify any built heritage within the assessment area based on examination on the following resources:

- List of declared and proposed monuments[1];
- List of graded historic buildings[2];
- List of new items for grading assessment[3];
- Government historic sites identified by AMO[4];
- Previous related EIA studies, publications and monographs on relevant historical and geographical issues;
- Unpublished archival papers and records, and collection and libraries of tertiary institutions; and
- Geological and historical maps, aerial photos and relevant visual archives.

12.3.3 Site visits were carried out within the assessment area on 20th January and 2nd June 2022, 21st December 2023 and 13th January 2025 to evaluate the current condition of the built heritage identified during the desktop review, as well as any items that might not be revealed by the desktop review.

12.3.4 The potential direct and indirect impacts on the built heritage during the construction and operational phases of the Project have been assessed in the CHIA by following the procedures and requirements of Annexes 10 and 19 of the EIAO-TM.

12.3.5 Mitigation measures, where necessary, would be recommended in the CHIA for any affected built heritage to minimise any adverse impacts.

Archaeological Impact Assessment

12.3.6 A desktop review was conducted to identify any potential existence of archaeological heritage within the assessment area based on examination on the following resources, when available:

[1] Antiquities and Monuments Office. *List of Declared Monuments and Proposed Monuments (as at 10 October 2024)*. Retrieved from https://www.amo.gov.hk/filemanager/amo/common/form/DM_Mon_List.pdf.
[2] Antiquities and Monuments Office. *List of Graded Historic Buildings (as at 26.6.2025)*. Retrieved from https://www.amo.gov.hk/filemanager/amo/common/form/List%20of%20Graded%20Historic%20Buildings.pdf.
[3] Antiquities Advisory Board. *List of New Items for Grading Assessment with Assessment Results (as at 3 September 2025)*. Retrieved from https://www.aab.gov.hk/filemanager/aab/en/content_29/list_new_items_assessed.pdf.
[4] Antiquities and Monuments Office. *Government Historic Sites Identified by AMO (as at May 2022)*. Retrieved from https://www.amo.gov.hk/filemanager/amo/common/form/build_hia_government_historic_sites.pdf.

- List of proposed and declared monuments[5];
- List of Sites of Archaeological Interest[6] identified by the AMO;
- Previous related EIA studies and archaeological reports;
- Related publications and monographs on relevant archaeological, historical and geographical issues;
- Unpublished archival papers and records, and collection and libraries of tertiary institutions; and
- Geological and historical maps, aerial photos and relevant visual archives.

12.3.7 Site visits were carried out on 20th January and 2nd June 2022, 21st December 2023 and 13th January 2025 within the assessment area to collate findings in desktop study and the on-site conditions of the identified archaeological potential areas within this Project. Photos taken during the site visits are provided in **Appendix 12.3**.

12.3.8 The potential impacts that may affect the possible archaeological heritage during the construction and operational phases have been assessed in the CHIA by following the procedures and requirements in Annexes 10 and 19 of the EIAO-TM.

12.3.9 In case direct impacts on archaeological heritage cannot be avoided during the early planning stage, justifications would be provided. Appropriate mitigation measures to minimise the impacts would be suggested for comment and agreement by AMO.

12.3.10 Part of the Project Site overlaps with the assessment areas[7] of the EIA studies of Northern Link (NOL) (Register No.: AEIAR-259/2024) (hereinafter referred to as "NOL Main Line EIA report") and San Tin / Lok Ma Chau Development Node (STLMC DN) (Register No.: AEIAR-261/2024) (hereinafter referred to as "STLMC DN EIA report"). The AIA for both studies had developed archaeological prediction models which were served as references for the purpose of archaeological potential analysis under the Project. As the archaeological resources would only be impacted by the proposed developments within the project areas of the two studies, the archaeological fieldworks and subsequent detailed analysis focused solely on their project areas[8], while there were no thorough analysis on the archaeological resources within areas outside the project areas but within the assessment areas. In view of this, further detailed analysis and in-depth study on the Project Site is deemed necessary.

12.4 Project Background

Geographical and Geological Background

12.4.1 The Project Site is located to the south of San Tin and to the northeast of Yuen Long. Notable village settlements include Yau Tam Mei Tsuen (攸潭美村), Wai Tsai Tsuen (圍仔村), Chuk Yuen (竹園), San Wai (新圍) and Long Ha (朗廈) which remain today are scattered around the Project Site. Tam Mei Barracks is situated

[5] Antiquities and Monuments Office. *List of Declared Monuments and Proposed Monuments (as at 10 October 2024)*. Retrieved from https://www.amo.gov.hk/filemanager/amo/common/form/DM_Mon_List.pdf.
[6] Antiquities and Monuments Office. *List of Sites of Archaeological Interest in Hong Kong (as at Nov 2012)*. Retrieved from https://www.amo.gov.hk/filemanager/amo/common/form/list_archaeolog_site_eng.pdf.
[7] 500 m assessment area refers to the area extending 500 m from the Project Boundary of STLMC DN and Proposed At-Grade Works Site and Works Area of NOL Main Line.
[8] Section 12.3.1.1 of the approved STLMC DN EIA report stated that "*the AIA is to assess the possible impact on any archaeological resources fall within the Project area*". Archaeological fieldworks and subsequent detailed analysis for both AIA studies under the STLMC DN and NOL Main Line EIA reports were only conducted within their project areas .

to the north, while Ngau Tam Mei Water Treatment Works (NTMWTW) is located immediately to the east. The northern and southern areas beyond Project Site are characterised by hilly and mountainous terrain.

Physical Geography and Past Landscape

12.4.2 Ngau Tam Mei (NTM) area is a valley located in between the north of Kai Kung Leng (雞公嶺) and the south of Ngau Tam Shan (牛潭山). The terrestrial elevation in this area ranges approximately from +2 mPD to +49 mPD. The majority of the land is flat, with gradients ranging roughly between <1° and 10°; while there are some small hill slopes to the north and south with gradients ranging roughly between 10° and 42°.

12.4.3 NTM area encompasses various landscapes, including river terraces, hill slopes, as well as coastal environment that presented before modern development. The Project Site is situated within the drainage basin of NTM area and extended towards the NTMWTW.

12.4.4 The volcanic bedrock (JTM, coarse ash crystal tuff) of Kai Kung Leng was formed during the Upper Jurassic period as part of the Tai Mo Shan Formation[9]. The hillslopes were formed by Pleistocene debris flow deposits (*Qpd*) overlaying the bedrock. During the Pleistocene, rivers running through the NTM valleys formed the current second river terrace, characterised by terraced alluvium deposits (*Qpa*). The current first river terrace was formed later during the Holocene, around 6,000BP[10], as a result of a drop in sea level. This is reflected by the alluvium deposits (*Qa*). The superficial deposits within the Project Site primarily consist of alluvial deposits (*Qa* and *Qpa*). Debris flow deposits (*Qpd*) can be found near the hillside in the southern parts. The western part comprises of estuarine deposits (*Qam*) and marine mud (*QHH*), indicating the pre-urbanised coastline and outer sea (**Figure 12.2** refers).

Historical Background

Qin (秦) to Yuan (元) Dynasties (221 BC to AD 1368)

12.4.5 Clues of human occupation within the southern China can be found in historic textual records such as *Shiji* (史記) and *Hanshu* (漢書) written in the first century BC to first century AD. These records describe that *Yue* (越) ethnic groups (also called *Hundreds of Yue* (百越)) were scattered around in southern China. Such ethnic groups comprised of different tribes bearing various surnames and could be differentiated from the *Han* (漢) ethnic group who lived in central China in terms of physical characteristics, language, and folklore.

12.4.6 The *Yue* people were gradually assimilated by the *Han* culture when southern China became an administration territory of China's central government since *Qin* dynasty (221 to 206 BC). During *Qin* period, the *Guangdong* (廣東) region was subordinated to *Panyu* (番禺) County. In 208 BC, *Southern Yue State* (南越國) was established in around the *Guangdong* region by military officials, who were sent from the *Qin* Court to conquer *Yue* in the south. Following the collapse of *Qin's* political power in the north, *Han* dynasty (206 BC to AD 220)[11] began. *Southern Yue State* soon became a vassal state of *Han* before being integrated into the *Han* Empire.

[9] Geotechnical Control Office. (1989). *Hong Kong Geological Survey Sheet 2*. Government of Hong Kong.
[10] Fyfe, J. A et al. (2000). *The Quaternary Geology of Hong Kong*. Hong Kong: Civil Engineering Department.
[11] 司馬遷 (c.a. 91BC)。史記 卷一百一十三 南越列傳 第五十三。北京：中華書局 (1959) 。

12.4.7 During *Han* to *Eastern Jin* (東晉) dynasties (AD 317 to 420), Hong Kong was subordinated to *Boluo* (博羅) County[12]. From AD 331 to AD 756, Hong Kong was subordinated to *Bao'an* (寶安) County. After AD 757, Hong Kong was subordinated to *Dongguan* (東莞) County and followed by *Song* (宋) dynasty (AD 960 to 1279) and *Yuan* dynasty (AD 1271 to 1368)[13].

Ming (明) to Qing (清) Dynasties (AD 1368 to 1912)

12.4.8 During the 15th century, the coastal areas of *Dongguan* County suffered from frequent marauding bandit and pirate attacks. *Xin'an* (新安) County was thus set up in AD 1573 to defend such attacks. According to *Xin'an Gazetteer* (新安縣誌)[14], the present-day New Territories, Kowloon and Hong Kong were zoned within *Xin'an* County.

12.4.9 In AD 1661, Coastal Evacuation Order was compelled by the *Qing* Court in order to stifle the anti-Manchu troops in Taiwan. People living in coastal area were forced to move 50 *li* (里) (approximately 25 km) inland, including the New Territories inhabitants. It was until AD 1669 people could move back to their villages. However, after the coastal evacuation, population severely dropped from 13,302 as recorded in AD 1594 to 3,912 in AD 1677. Thus, *Hakka* (客家) people were encouraged to move into the New Territories area during late 17th to early 18th centuries[15].

12.4.10 After the First Opium War (1839 to 1842) between the *Qing* government and the British Empire, the *Qing* government ceded the Island of Hong Kong to the Great Britain under the *Treaty of Peace, Friendship, Commerce, Indemnity, etc., Between Great Britain and China* (also known as the *Treaty of Nanking* (南京條約))[16], which was signed in 1842. The *Qing* government lost the Second Opium War (1856 to 1860), which led to the ceding of Kowloon as a dependency of Hong Kong under the *Convention of Peace and Friendship Between Great Britain and China*[17] (also known as the *Convention of Peking* (北京條約)) in 1860. At the turn of the 20th century, *the Convention between Great Britain and China Respecting an Extension of Hong Kong Territory*[18] (展拓香港界址專條) (also known as the *Second Convention of Peking* (第二北京條約)) signed between the British and the *Qing* government in 1898 allowed the British colony to enlarge the limits of British territory under lease for ninety-nine years.

Modern Period (after 1912)

12.4.11 No major historic event took place in the NTM area in the 20th century until the World War II. Castle Peak Road which was constructed between 1911 and 1920 serves as one of the main roads that passes through the NTM area, connecting Sham Shui Po, Tuen Mun, Yuen Long and Sheung Shui. Aerial photos between

[12] Although the boundary between *Boluo* (博羅) County and *Panyu* (番禺) County during *Han* to East *Jin* period is unclear, it is generally suggested that Hong Kong region belonged to *Boluo* County at that time, according to *Xinan Gazetteer* (1819), Social Change in Hong Kong Before and After the Early Qing Clearance (1986), and Brief History of Ancient Shenzhen (1997). However, Professor Jao Tsung-I (2005) discussed that the area belonged to *Panyu* based on the inscriptions on bricks of Lei Cheng Uk Han Tomb.
[13] 劉智鵬、劉蜀永 (編) (2020)。《方志中的古代香港-《新安縣志》香港史料選》。香港：三聯書店（香港）有限公司。
[14] *Ibid*.
[15] 蕭國健 (1986)。《清初遷界前後香港之社會變遷》。台北：台灣商務印書館。
[16] Hong Kong Government. (1964). *Laws of Hong Kong. Appendix IV A Selection of Constitutional Documents, Conventions and Treaties*. Retrieved from https://oelawhk.lib.hku.hk/archive/files/87f64e08aedf20f07b7d8ac2132030c1.pdf.
[17] *Ibid.*
[18] *Ibid.*

1924 and 1956 show that there was no significant change to the settlement patterns and agricultural land use[19].

12.4.12 The Japanese troops invaded Hong Kong on 8th December 1941 and had travelled via Castle Peak Road from Lok Ma Chau to Yuen Long and Kowloon. They passed through NTM area from 8th to 9th December 1941[20].

12.4.13 The industrialisation of Hong Kong and imports of cheap food from Mainland China in the 1950s and 1960s had influenced the key economic activities in the Yuen Long area, changing from rice cultivation to vegetation cultivation[21]. Agricultural fields in NTM area were gradually replaced by fishponds since the late 1950s[22] and 1960s[23] [24], while the hill lands owned by the *Man* (文) clan were rented to *Chiu Chau* (潮州) immigrants who then successfully cultivated vegetables. Houses were also gradually seen occupying the once agricultural fields in the plains in NTM valley during the 1960s[25] (**Figures 12.3** and **12.4** refer).

12.4.14 Construction of San Tin Highway commenced in the late 1980s as part of Route 9 in Hong Kong (also known as New Territories Circular Road). Simultaneously, a series of fishponds were filled in the area. The agricultural landscape in the NTM area has been largely transformed into factories, temporary storage and open carpark areas upon the completion of San Tin Highway.

12.4.15 A series of civil engineering works were carried out since 1990s to improve the living environment in the NTM area, such as construction of the channelisation of NTM Drainage Channel between 2001 and 2003[26], and the establishment of NTMWTW in 2000[27].

Village Development

12.4.16 Most of the villages in the NTM area are located at the mouth of the valley or in close proximity to coastal settings. The assessment area comprises of five villages, namely (i) Yau Tam Mei Tsuen, (ii) Wai Tsai Tsuen, (iii) Chuk Yuen, (iv) San Wai and (v) Long Ha.

Yau Tam Mei Tsuen (Appendix 12.1, Built Heritage and Other Identified Items Sheet YTMT refers)

12.4.17 Yau Tam Mei Tsuen is situated within the NTM valley, with settlements scattered along the river and foothills in the area.

12.4.18 Yau Tam Mei Tsuen was originally named Ngau Tam Mei Tsuen (牛潭尾村) (literally *a village at the cattle back's pool*). However, due to the distaste for the words "牛" (*cattle*), it was later renamed to Yau Tam Mei Tsuen ("攸潭美村", literally translated as *a village with a beautiful steady flowing water pool*). The majority of

[19] Lands Department. (1924). *1:14332, ---ft., H32-0013 [aerial photo].* Retrieved from https://www.hkmapservice.gov.hk/OneStopSystem/map-search.
Lands Department. (1924). *1:13800, 11500ft., H8-0005 [aerial photo]*. Retrieved from https://www.hkmapservice.gov.hk/OneStopSystem/map-search.
Lands Department. (1945). *1:12000, 20000ft., 681_4-4138 [aerial photo]*. Retrieved from https://www.hkmapservice.gov.hk/OneStopSystem/map-search.
[20] History in Data Project website (Accessed on 11 March 2022) Hong Kong 1941. Retrieved from https://digital.lib.hkbu.edu.hk/1941hkbattle/en/research-team.php.
[21] Johnson, E. (2000). *Recording a rich heritage: research on Hong Kong's "New Territories"*. Hong Kong: Leisure & Cultural Services Department.
[22] Lands Department. (1956). *1:10020, 16700 ft. F22_560-0205 [aerial photo]*. Lands Department.
[23] Lands Department. (1963). *1:27000, 13500 ft., V81A_857-0013R [aerial photo]*. Lands Department.
[24] Lands Department. (1975). *1:25000, 12500ft, 11882R [aerial photo]*. Lands Department.
[25] Lands Department. (1964). *1:25000, 12500 ft., 1964-2790R [aerial photo]*. Lands Department.
[26] Lands Department. (2006). *1:5 000, 2-SE-C (Ed 2006-08) [topographic map]*. Lands Department.
[27] Water Services Department. *Ngau Tam Mei Water Treatment Works*. Retrieved from https://www.wsd.gov.hk/filemanager/en/share/pdf/NgauTamMeiTW-e.pdf.

the land in the NTM area was initially owned by the *Man* clan from San Tin who engaged in agricultural activities. The settlers in the NTM area had to either rent or purchase land from them.

12.4.19 The construction year of Wai Cheung Ancestral Hall can be dated back to 1887 (丁亥) (**Sections 12.5.12** to **12.5.13** refer), suggesting that people had already settled in the western part of the NTM area since the late 19**th** century. Most of the settlers in Yau Tam Mei Tsuen were *Hakka* people from Mainland China.

12.4.20 Yau Tam Mei Tsuen in general is a multi-clan village, with the majority of residents belong to the *Lees* (李), *Chow* (周), *Chu* (朱), *Yeung* (楊), *Cheung* (張), and *Lau* (劉) clans.

Wai Tsai Tsuen (Appendix 12.1 Built Heritage and Other Identified Items Sheet WTT refers)

12.4.21 Wai Tsai Tsuen is situated to the north of the mouth of the NTM river valley. It was originally established by the *Wens* (尹) from *Dongguan* during the *Qing* dynasty (1636 – 1912). The village was initially located elsewhere, but its occupants relocated to the current village area due to development following the British colonisation of the New Territories in 1898.

12.4.22 Originally named Tung Hing Tsuen (東慶村), it was later renamed as Wai Tsai. The *Cheungs* from *Waiyang* (惠陽), *Lungkong* (龍岡), and *Pingshan* (坪山) settled in the village after the Japanese occupation.

Chuk Yuen (Appendix 12.1, Built Heritage and Other Identified Items Sheet CY refers)

12.4.23 Chuk Yuen is located east of Castle Peak Road and south of Chuk Yau Road. It was established in the 17th century by the *Wongs* (黃), who originated from *Dongguan*. The village's presence was recorded in the *Xin'an Gazetteer* as "Chuk Yuen Wai" (竹園圍) in the version edited during the 27th year of the *Kangxi* (康熙) reign (1688) and the 24th year of the *Jiaqing* (嘉慶) reign (1819). Throughout its history, Chuk Yuen has been home to various clans, including the *Lees*, *Chows*, *Yuens* (袁), and *Aus* (區).

12.4.24 Sheung Chuk Yuen (上竹園) was established in the eastern part, while Ha Chuk Yuen (下竹園), now known as Chuk Yuen Tsuen (竹園村), was developed in the western part. Ha Chuk Yuen is the older of the two villages, while Sheung Chuk Yuen is comparatively newer. Castle Peak Road and San Tin Highway serve as physical dividers between Sheung Chuk Yuen and Ha Chuk Yuen.

San Wai (Appendix 12.1, Built Heritage and Other Identified Items Sheet SW refers)

12.4.25 San Wai is located east of Castle Peak Road and south of the NTM river valley. Originally known as *Fuk Hing Lane* (福慶里), the village was initially inhabited by the *Wongs* and *Chius* (趙) in the early 20th century. However, it was subsequently abandoned for a period of time.

12.4.26 During the late *Qing* and early Republican period, the village became home to the *Yeungs* (楊). It was during this time that the village was renamed San Wai or San Wai Tsai. Later, the *Lees*, *Wongs* and *Chans* (陳) also settled in the village. San Wai was occupied by Japanese troops during the Japanese occupation era.

12.4.27 As the population grew in the early 20th century, new villages were established. Ha San Wai became the older village in the west, while Sheung San Wai emerged as the newer village in the east. Today, Ha San Wai and Sheung San Wai are known

as Ha San Wai (west) and San Wai (east), separated by Castle Peak Road and the San Tin Highway.

Long Ha (Appendix 12.1, Built Heritage and Other Identified Items Sheet LH refers)

12.4.28 Long Ha is situated to the south of San Wai and occupies the eastern foothill area of Kai Kung Leng. A small mound is located at the eastern end of a local road that connects to the San Tin Highway to the west**.** It is likely that Long Ha has historical ties to this particular branch of the *Chan* clan from *Taishan* (台山) and *Yingchuan* (潁川).

Archaeological Background

12.4.29 There is one Site of Archaeological Interest (SAI), namely Ngau Tam Mei SAI (SAI38) [28] is located partially within the Project Site. In addition, previous archaeological investigations and assessments were conducted within the assessment area of the Project. The details are described as follows and their locations can be referred to **Figures 12.5** and **12.6**.

Ngau Tam Mei SAI (SAI38)[29]

12.4.30 During the 1997 Territorial-Wide Archaeological Survey in Yuen Long, significant findings were reported at the site. Pottery shards featuring distinctive "Union Jack" patterns, which have been dated to the Bronze Age, as well as tiles from the *Han* dynasty were found. These findings suggested human habitation in the area from the Bronze Age to the early *Han* dynasty. The site was later recognised as Ngau Tam Mei SAI [30] (**Figure 12.5** refers).

Archaeological Investigation Report on The Main Drainage Channels for Ngau Tam Mei Project, Phase II[31]

12.4.31 An archaeological survey was conducted in 1999 for the construction of channelised nullah in NTM[32]. Twenty-two augur holes and one trench were dug within the study area of the nullah project. Neither archaeological features nor artefacts were discovered.

12.4.32 In addition, this investigation in 1999 also carried out twelve augur holes tests and three trenches in the vicinity of the Ngau Tam Mei SAI to demark the boundary of the SAI (**Figure 12.5** refers). Pottery sherds dated to the late Bronze Age and tiles dated to the *Han* dynasty were found on the surface of the SAI, but no relics were discovered from the augur holes and trenches. This suggested that the archaeological potential is unlikely to spread into the study area of the nullah project.

EIA and Traffic Impact Assessment (TIA) Studies for the Stage 2 of PWP Item No. 215DS – Yuen Long and Kam Tin Sewerage and Sewage Disposal – Environmental Impact Assessment Report (Register No.: AEIAR-078/2004)[33]

[28] Antiquities and Monuments Office. *List of Sites of Archaeological Interest in Hong Kong (as of Nov 2012)*. Retrieved from https://www.amo.gov.hk/filemanager/amo/common/form/list_archaeolog_site_eng.pdf.
[29] 古物古蹟辦事處 (1997)。《全港文物普查 1997 第一地區(元朗區)工作報告》。香港：古物古蹟辦事處檔案。
[30] *Ibid*.
[31] 古物古蹟辦事處 (1999)。《元朗牛潭尾主渠道第二期工程考古調查工作報告》。香港：古物古蹟辦事處檔案。
[32] *Ibid*.
[33] Drainage Services Department. (2013). *EIA and TIA Studies for the Stage 2 of PWP Item No. 215DS – Yuen Long and Kan Tin Sewerage and Sewage Disposal (Agreement No. CE 66/2001). Section 15 Cultural Heritage Assessment.* Retrieved from https://www.epd.gov.hk/eia/files/applications/en/pp_82/eia_1578/progress/action_3229/eia_report.pdf.

12.4.33 An AIA was conducted for the construction of Yuen Long and Kam Tin Sewerage and Sewage Disposal system in 2004. Sewer alignment from Kam Tin River to NTM sewage pumping station (SPS), NTM SPS, sewer alignment from NTM SPS to Tam Mei Camp SPS, Tam Mei Camp SPS, and sewer alignment from NTM SPS to San Tin Camp SPS were assessed of their archaeological potential (**Figure 12.6.1** refers). The area was scattered with former ponds, estuarine sediments, river meandering and road with cumulative impacts of utilities. Thus, there is no archaeological potential.

Environmental Impact Assessment Study for Northern Link – Environmental Impact Assessment Report (Register No.: AEIAR-259/2024)[34]

12.4.34 An archaeological survey was conducted in 2023 presented in the NOL Main Line EIA report. Three test pits along with field scanning were carried out in NTM area, which are located within the Project Site. In addition, field scanning was conducted at Long Ha Tsuen, which is located within the 300 m assessment area but falls outside the Project Site (**Figure 12.5** refers). Neither archaeological features nor artefacts were discovered in the surveyed area in NTM area.

12.4.35 Based on the survey results, the surveyed area in NTM would have possess low archaeological potential. For the surveyed area at Long Ha Tsuen, it was suspected to have experienced large scale of site formation works, soil removal and slope cutting were observed. Therefore, with evaluation towards its past land use, it was confirmed that the surveyed area does not contain any archaeological potential.

12.4.36 The archaeological potential within assessment area in the NOL Main Line EIA report (**Figure 12.6.2** refers) was also assessed based on the desktop study, archaeological survey and the results of archaeological predictive models. Among the twenty-three areas of archaeological potential identified within the assessment area in the NOL Main Line EIA report, five of them are located within the 300 m assessment area of this Project, namely (i) Chuk Yuen Tsuen Area, (ii) Long Ha Area, (iii) Ngau Tam Mei Area, (iv) Ngau Tam Mei (North) Area and (v) Siu Hum Tsuen Area. In particular, Ngau Tam Mei Area and Ngau Tam Mei (North) Area are located within the Project Site.

First Phase Development of the New Territories North – San Tin / Lok Ma Chau Development Node – Investigation – Environmental Impact Assessment Report (Register No.: AEIAR-261/2024)[35]

12.4.37 An archaeological survey was conducted for STLMC DN, including field scanning of about 52 hectares of land and two test pit excavations. In particular, field scanning area A9 located near Shek Wu Wai San Tsuen has encroached into the assessment area of this Project, but outside the Project Site (**Figure 12.5** refers). It was observed that A9 contained many modern trashes, and some signs of site flattening in modern times that would have already disturbed the ground. As no archaeological material was identified, the area was considered to have no archaeological potential.

[34] MTR Corporation Limited. (2023). *Environmental Impact Assessment Study for Northern Link (Agreement No. C1603). Section 13 Cultural Heritage Impact Assessment.* Retrieved from https://www.epd.gov.hk/eia/files/applications/en/pp_346/eia_1992/progress/action_1365/eia_3012023/HTML/EIA%20Report/S13%20CHIA.htm

[35] Civil Engineering and Development Department and Planning Department. (2023). *First Phase Development of the New Territories North – San Tin / Lok Ma Chau Development Node – Investigation (Agreement No. CE 20/2021). Section 12 Impact on Cultural Heritage.* Retrieved from https://www.epd.gov.hk/eia/files/applications/en/pp_340/eia_1994/progress/action_157674/eia_3022023/EIA/CE20_EIA_S12.htm#_Toc154070604.

12.4.38 Based on the desktop analysis, archaeological fieldworks and archaeological predictive modelling, the archaeological potential within the assessment area in the STLMC DN EIA report was assessed (**Figure 12.6.3** refers). Among the thirteen areas of moderate and high archaeological potential identified within the assessment area in the STLMC DN EIA report, Ngau Tam Mei (North) Area has encroached into this Project Site, while Siu Hum Tsuen Area falls within the 300 m assessment area but outside the Project Site.

Ngau Tam Mei Water Treatment Works Extension – Environmental Impact Assessment Report (Register No.: AEIAR-262/2024)[36]

12.4.39 A desktop review was conducted for the extension of Ngau Tam Mei Water Treatment Works (NTMWTW). The proposed works do not encroach into the Project Site, but fall within the 300 m assessment area (**Figure 12.6.4** refers). Based on the evaluation of the area's geological conditions, the proposed works are either located on hilly landscape that is unfavourable for human settlement, or existing road network, disturbed areas with existing utilities and cut slope. As such, these areas are considered to have no archaeological potential.

12.5 Built Heritage Impact Assessment

Baseline Condition

12.5.1 This section summarises built heritage identified within the assessment area according to the definition and categorisation discussed in the following sections.

12.5.2 For the purpose of this assessment, both known built heritage and unknown built heritage are defined as follows:

12.5.3 Known built heritage are the tangibles (buildings / structures / features / places / areas) that have been identified and / or accorded with a status. They are either under statutory protection or administrative protection. The known built heritage exclusively includes the following:

i. **declared monuments under A&MO**: the concerned place, building, site or structure which is declared to be a monument, historical building or archaeological or palaeontological site or structure under section 3 of the A&MO (Cap. 53); and included in the latest list of Declared Monuments in Hong Kong.

ii. **proposed monuments under A&MO**: the concerned place, building, site or structure which is declared to be a proposed monument, proposed historical building, or proposed archaeological or palaeontological site or structure under section 2A of the A&MO (Cap. 53).

iii. **grade 1 historic buildings assessed by the** Antiquities Advisory Board (**AAB**): as defined by the AAB, they are buildings of outstanding merit, which every effort should be made to preserve if possible.

iv. **grade 2 historic buildings assessed by the AAB**: as defined by the AAB, they are buildings of special merit; efforts should be made to selectively preserve.

v. **grade 3 historic buildings assessed by the AAB**: as defined by the AAB, they are buildings of some merit; preservation in some form would be

[36] Water Supplies Department. (2024). *Ngau Tam Mei Water Treatment Works Extension – Environmental Impact Assessment Report (282551-REP-080-05). Section 11 Impact on Cultural Heritage.* Retrieved from https://www.epd.gov.hk/eia/register/study/latest/esb-333.pdf.

desirable and alternative means should be considered if preservation is not practicable.

vi. **new items pending for grading assessment by the AAB**: currently under the latest *list of new items for grading assessment with grading results* pending for grading assessment by the AAB, and yet to be grading accorded.

vii. **government historic site identified by the AMO**: under the list of government historic site maintained by the AMO.

12.5.4 Unknown built heritage are the tangibles (buildings / structures / features / places / areas) that have not been identified and / or accorded with a status. They are not included under the known built heritage listed above. They would be assessing from perspectives of historical, architectural, cultural as well as group values[37]. Particular items would also be considered should they are excluded from the known built heritage listed above.

12.5.5 For the purpose of this assessment, known built heritage identified are presented as "built heritage" and unknown built heritage identified are presented as "other identified items" in this Report.

Built Heritage and Other Identified Items Within the Assessment Area

12.5.6 No built heritage, including declared monuments, proposed monuments, graded historic buildings or government historic sites identified by the AMO, is identified within the Project Site. Nine other identified items are located within the Project Site.

12.5.7 There is a total of forty-five built heritage and other identified items located outside the Project Site but within the 300 m assessment area, including one grade 2 historic building; nine grade 3 historic buildings; and thirty-five other identified items.

12.5.8 A summary of the built heritage and other identified items is presented in **Table 12.1**. Their locations are presented in **Figure 12.7**.

12.5.9 Each of the built heritage and other identified item has been given a reference number. For any buildings / structures that has been accorded with an official status or included under grading assessment by AAB, it would be numbered based on its status (e.g. declared monument (DM) and historic building (HB)) and the number given by the respective authorities.

12.5.10 For other identified items that are not previously included under grading assessment by AAB, they would be numbered based on the abbreviation of village names (i.e. Yau Tam Mei Tsuen, Wai Tsai Tsuen, Chuk Yuen, San Wai and Long Ha) or the area where the building / structure is located (i.e. General (NTM Area)).

[37] Australia International Council and Monuments and Sites (ICOMOS). (2013). *The Burra Charter: the Australia ICOMOS Charter for Place of Cultural Significance*. Retrieved from https://australia.icomos.org/wp-content/uploads/The-Burra-Charter-2013-Adopted-31.10.2013.pdf.

Table 12.1 Built Heritage and Other Identified Items within the Assessment Area

Ref. No.	Name	Status	Approx. Distance to Project Site	Details and Photographic Record	Relevant Figure No.
Yau Tam Mei Tsuen					
HB1219	Wai Cheung Ancestral Hall	AAB discussed in 2010 and confirmed as no grading	Within Project Site	**Appendix 12.1**, Built Heritage and Other Identified Items Recording Sheet YTMT (Yau Tam Mei Tsuen)	12.7.4
YTMT01	DD104 Lot 4186 S.E (Residence)	No Status	Within Project Site		12.7.9
YTMT02	DD104 Lot 4187 S.B (Watchtower)	No Status	Within Project Site		12.7.9
YTMT03	Remnants of Nam Shan Monastery	No Status	Within Project Site		12.7.9
YTMT04	Subsidiary Station of San Yau Vegetable Marketing Co-operative Society, Ltd.	No Status	Within Project Site		12.7.10
YTMT05	San Yau Vegetable Marketing Co-operative Society, Ltd.	No Status	6 m		12.7.5
YTMT06	Former Yau Tam Mei Primary School	No Status	Within Project Site		12.7.10
YTMT07	Lee's Boundary Stone	No Status	Within Project Site		12.7.9
Wai Tsai Tsuen					
WTT01	DD104 Lot 2729 (Residence)	No Status	25 m	**Appendix 12.1**, Built Heritage and Other Identified Items Recording Sheet WTT (Wai Tsai Tsuen)	12.7.1 & 12.7.2
WTT02	Nos. 9-11, Wai Tsai Tsuen	No Status	82 m		12.7.2
WTT03	Nos. 13-15, Wai Tsai Tsuen	No Status	93 m		12.7.2
WTT04	No. 44, Wai Tsai Tsuen	No Status	99 m		12.7.2
WTT05	No. 47, Wai Tsai Tsuen	No Status	107 m		12.7.2
WTT06	No. 51, Wai Tsai Tsuen	No Status	107 m		12.7.2
WTT07	No. 54, Wai Tsai Tsuen	No Status	111 m		12.7.2
WTT08	Former Kai Tak School	No Status	97 m		12.7.2
Chuk Yuen					
HB1383	Kong Ha Tong, No. 34, Sheung Chuk Yuen	AAB discussed in 2010 and confirmed as no grading	149 m	**Appendix 12.1**, Built Heritage and Other	12.7.5

Ref. No.	Name	Status	Approx. Distance to Project Site	Details and Photographic Record	Relevant Figure No.
HB1384	Kong Ha Tong, No. 35, Sheung Chuk Yuen	AAB discussed in 2010 and confirmed as no grading	152 m	Identified Items Recording Sheet CY (Chuk Yuen)	12.7.5
CY01	No. 3, Sheung Chuk Yuen	No Status	83 m		12.7.5
CY02	No. 37, Sheung Chuk Yuen	No Status	168 m		12.7.5
CY03	No. 35, Chuk Yuen Tsuen	No Status	93 m		12.7.3
CY04	No. 72, Chuk Yuen Tsuen	No Status	128 m		12.7.3
CY05	Nos. 74-75, Chuk Yuen Tsuen	No Status	136 m		12.7.3
CY06	Temple, Chuk Yuen Tsuen	No Status	168 m		12.7.3
CY07	Chuk Hing Public School	No Status	115 m		12.7.3
San Wai					
HB376	No. 57, San Wai Tsuen	Grade 2 Historic Building	195 m	**Appendix 12.1**, Built Heritage and Other Identified Items Recording Sheet SW (San Wai)	12.7.5 & 12.7.7
HB613	No. 50, San Wai Tsuen	Grade 3 Historic Building	212 m		12.7.5 & 12.7.7
HB628	No. 51, San Wai Tsuen	Grade 3 Historic Building	218 m		12.7.5 & 12.7.7
HB658	No. 35, San Wai Tsuen	Grade 3 Historic Building	57 m		12.7.6
HB659	No. 36, San Wai Tsuen	Grade 3 Historic Building	58 m		12.7.6
HB784	No. 71, San Wai Tsuen	Grade 3 Historic Building	251 m		12.7.7
HB785	No. 87, San Wai Tsuen	Grade 3 Historic Building	62 m		12.7.6
HB858	No. 70, San Wai Tsuen	Grade 3 Historic Building	209 m		12.7.7
HB1016	Hon Lo, No. 61, San Wai Tsuen	Grade 3 Historic Building	163 m		12.7.7
HB1032	No. 62, San Wai Tsuen	Grade 3 Historic Building	175 m		12.7.7
SW01	No. 3, San Wai Tsuen	No Status	64 m		12.7.6
SW02	No. 10, San Wai Tsuen	No Status	52 m		12.7.6
SW03	Nos. 16-17, San Wai Tsuen	No Status	48 m		12.7.6
SW04	No. 25A, San Wai Tsuen	No Status	20 m		12.7.6
SW05	Nos. 39-40, San Wai Tsuen	No Status	177 m		12.7.5
SW06	Nos. 46-47, San Wai Tsuen	No Status	183 m		12.7.5
SW07	No. 54, San Wai Tsuen	No Status	172 m		12.7.5 & 12.7.7
SW08	No. 59, San Wai Tsuen	No Status	217 m		12.7.5 & 12.7.7

Ref. No.	Name	Status	Approx. Distance to Project Site	Details and Photographic Record	Relevant Figure No.
SW09	No. 67 and Sin Lo, San Wai Tsuen	No Status	216 m		12.7.7
Long Ha					
LH01	Chan Ming Yam Tong	No Status	209 m	**Appendix 12.1**, Built Heritage and Other Identified Items Recording Sheet LH (Long Ha)	12.7.8
General (NTM Area)					
NB01	Grave of Mrs Man Ng	No Status	63 m	**Appendix 12.1**, Built Heritage and Other Identified Items Recording Sheet NB (General (NTM Area))	12.7.11
NB02	Mailbox No. 35	No Status	13 m		12.7.6
NB03	Mailbox No. 169	No Status	37 m		12.7.1
NB04	Mailbox No. 299	No Status	Within Project Site		12.7.3
NB05	Man's Boundary Stone	No Status	40 m		12.7.11
NB06	Castle Peak Road Milestone	No Status	52 m		12.7.3
NB07	Tam Mei Barracks	No Status	Immediately outside the Project Site		12.7
NB08	Chun Chi Education Park	No Status	33 m		12.7.1
NB09	Grave of Mr. Man Chiu Pak and His Wife	No Status	Within Project Site		12.7.10

12.5.11 There is no built heritage within the Project Site. The other identified items located within the Project Site and their key significance are discussed in the following sections. The remaining built heritage and other identified items located within the assessment area but outside Project Site, along with those mentioned below, are presented in the Built Heritage and Other Identified Items Recording Sheets in **Appendix 12.1** with details and photo records.

*Yau Tam Mei Tsuen (**Appendix 12.1**, Built Heritage and Other Identified Items Recording Sheet YTMT refers)*

- *Wai Cheung Ancestral Hall (HB1219)*

12.5.12 Wai Cheung Ancestral Hall (維祥公祠) was constructed in 1887 (丁亥) by *Chow Yam* (周壬) and his brother *Chow Lung* (周龍) of the *Chow* clan. The ancestral hall is one of the few remaining examples of Chinese vernacular buildings made of grey bricks in Yau Tam Mei Tsuen. Initially, the ancestral hall was a three-bay one-hall building constructed in *Qing* (清) vernacular style. In the 1950s, additional bay was added on each side of the ancestral hall to serve as residences for the *Chows*, each with its own entrance. It features grey brick and mud brick construction, with a pitched clay-tiled roof supported by timber rafters and purlins. The top of the

front wall of the central bay is decorated with wall frieze paintings of flowers, birds, and calligraphy. Additionally, a ceramic cock decoration is placed at the centre of the ridge, facing east.

12.5.13 Wai Cheung Ancestral Hall holds significant historical, architectural and cultural significance. Being the few surviving grey brick ancestral hall in NTM area, it witnesses the settlement of the *Chow* clan since the late 19th century. The construction of ancestral hall by the *Chow* clan also reflects the traditional ancestral worship practices in Chinese culture.

- *DD104 Lot 4186 S.E (Residence) (YTMT01)*

12.5.14 The house in NTM is a single bay flat roof structure constructed using concrete. It follows a rectangular plan and has two storeys. The front façade is rendered with plaster. Above the entrance, the Chinese characters "何生農場", meaning *Ho's* Farm, are painted.

12.5.15 The significance in cultural heritage of the house is noteworthy in terms of historical and architectural perspective. It provides insights into the past and serves as a tangible representation of the local community's livelihood in NTM and engagement in agricultural pursuits. Its architecture also demonstrates typical design of modern village house in the mid-20th century Hong Kong.

- *DD104 Lot 4187 S.B (Watchtower) (YTMT02)*

12.5.16 The watchtower is a one-hall structure with a pitched roof situated southwest of a row of village houses. The front façade features a plaque and couplet, although the characters have faded. The air opening is framed by a radiating sun design reminiscent of the twelve rays white sun associated with the Emblem of the Chinese Nationalist Party, *Kuomintang* (國民黨).

12.5.17 The watchtower holds architectural merit as it serves as a physical testament to the potential past presence of the Nationalist Party in this area. The closest indication of the Nationalist Party's presence can be found in the General House in Fung Kat Heung, which comprises three grade 2 historic buildings. These structures collectively provide insights into the Nationalist Party's influence and possible activities in the 20th century.

- *Remnants of Nam Shan Monastery (YTMT03)*

12.5.18 The remnants of Nam Shan Monastery consist of two gates and a pavilion. Located along a staircase, these three structures are primarily constructed using concrete. The Gate of To Tak of Nam Shan Monastery is situated by the roadside in the south of Yau Tam Mei Tsuen. The name of the entrance, "道德之門" (Entrance of Morality), is embossed on the plaque above the eaves. After the Gate of To Tak is the Gate of Nam Shan Monastery which bears the Chinese characters "南山精舍" (Nam Shan Monastery) embossed under the eaves. Further up the steps, there is a pavilion with a hexagonal plan. Each corner of the pavilion features a round column supporting the roof. The cornice under the roof is moulded with geometric shapes or a serrated pattern known as "牙子砌" (resembling teeth).

12.5.19 The cultural heritage significance of these monastic structures lies in their historical and social merit. While there is limited information available on the history of the monastery, they serve as tangible reminders of the religious history and practices that were once prominent and benefited the settlers in the region.

- *Subsidiary Station of San Yau Vegetable Marketing Co-operative Society, Ltd. (YTMT04)*

12.5.20 The San Yau Vegetable Marketing Co-operative Society, Ltd. (新攸蔬菜產銷有限責任合作社) was established on November 18, 1953. The subsidiary station which first appeared in 1963, initially consisted of a single storey building with a sheltered front porch. The existing structure was likely rebuilt in the mid-1970s. It follows a utilitarian style, which is constructed of reinforced concrete and features a single storey, a flat roof, and a covered front porch.

12.5.21 The significance in cultural heritage of the subsidiary station lies in its historical value. Along with other Vegetable Marketing Co-operative Societies in Hong Kong, these structures serve as a testament to the government's efforts to improve the agricultural industry and promote agricultural cooperative movements among farmers in the New Territories during the latter half of the 20th century. It also has group value with San Yau Vegetable Marketing Co-operative Society, Ltd. (YTMT05), which reflect the thriving vegetable farming in the NTM area.

- *Former Yau Tam Mei Primary School (YTMT06)*

12.5.22 Former Yau Tam Mei Primary School (前攸潭美小學) is a school complex comprises of three buildings, a basketball court, a latrine and two annexes. The school was initially established in 1931 as a private school by *Chu Shau-ki* (朱壽棋), named Tak San School (德新學校). Later, in 1939, it was re-established as Yau Tam Mei School. By 1946, the school had received government subsidies and its name was changed to Yau Tam Mei Primary School to signify its government affiliation.

12.5.23 The three school buildings share similar architectural style. They are all single-storey buildings in a generally rectangular plan. Their metallic gable roofs are supported by the concrete walls. One notable feature is the parapet of the main school building has adorned with the school's name, donor information, and likely the construction year of the main building in 1961.

12.5.24 The significance of cultural heritage of former Yau Tam Mei Primary School lies in its historical and social value. The history of the school was in line with the history of Hong Kong. The historical influx of Chinese refugees in the 1920s and 30s would have created the demand on the local need to provide education. On the other hand, the recommendation made to the government by Edmund Burney, Inspectors of Schools of England, in the 1930s after his visit in Hong Kong would have allowed the government to encourage local education furthermore in the subsequent years. Furthermore, the school has served the pupils of NTM for decades and played a significant role in education of the area. Architecturally, the school buildings also demonstrated a typical village school design in the New Territories.

- *Lee's Boundary Stone (YTMT07)*

12.5.25 The Lee's Boundary Stone is a carved stone situated on a hillslope in the southern part of the NTM valley. The southwest-facing side of the stone bears the Chinese character "李" (*Lee*). The exact dating of the stone is currently unknown.

12.5.26 The cultural heritage significance of this stone lies in its historical value as a witness to the settlement history of the *Lee* clan in NTM valley. It serves as a tangible link to the past, providing insights into the presence and influence of the *Lee* clan in the area.

*General (NTM Area) (**Appendix 12.1**, Built Heritage and Other Identified Items Recording Sheet NB refers)*

- *Mailbox No. 299 (NB04)*

12.5.27 Three mailboxes (Mailbox Nos. 35, 169 and 299) were identified along the paths of San Tin Highway and San Tam Road. In particular, mailbox No. 299 can be found near the junction of Chuk Yau Road outside Sheung Chuk Yuen within the Project Site. These mailboxes exhibit a similar design with the royal insignia "ERII" (Elizabeth Regina II) embossed at the bottom. The presence of this insignia indicates that these mailboxes belong to the era of Queen Elizabeth II's reign during the colonial period of Hong Kong.

12.5.28 Mailbox No. 299 serves as a witness to the colonial rule of the British government in Hong Kong. It holds both historical and social significance as it has played a role in serving the local community. Along with Mailbox Nos. 35 and 169 that being established at the same period, it also holds group value as a testament to their shared historical context.

- *Grave of Mr. Man Chiu Pak and His Wife (NB09)*

12.5.29 The Grave belongs to Mr. *Man Chiu Pak* (“文肖白”) in the *Man* clan and his wife, Ms. *Cheng* (“鄭氏”). He held a *Qing* government rank of *Gentleman for Cultivating Duty* (“修職郎”), at a Senior Eighth Rank (“正八品”), but the official duty of the deceased was not known from the grave tablet. The design of the grave follows traditional Chinese grave architecture. According to the inscriptions on the tablet, the grave was erected in the second year of the *Xuantong* reign of the *Qing* dynasty (1911). It is renowned as one of the prominent *fung shui* spots of the *Man* clan, known as *Fung Chui Lo Dai* (“風吹羅帶”), which translates to “the wind blows the silken sash”.

12.5.30 The cultural heritage significance of the grave is noteworthy from historical and cultural perspective. It serves as evidence of the *Man* clan's presence in the NTM area where the land was initially owned by the *Man* clan before rented to the settlers. Furthermore, the grave holds group value in conjunction with the Grave of Mrs Man Ng (NB01) and Man’s Boundary Stone (NB05), which mark the extent of the *Man* clan's influence in San Tin.

Identification and Evaluation of Impacts

12.5.31 A total of fifty-four built heritage and other identified items are identified within the 300 m assessment area. Nine of them are located within the Project Site, while the remaining forty-five are located outside the Project Site but within 300 m assessment area (**Table 12.1** refers).

12.5.32 Further assessment on Tam Mei Barracks (NB07) is not available due to restricted access. Noting that this Project would not directly alter the Barracks, it is thus not considered in this impact assessment.

Construction Phase

Built Heritage and Other Identified Items within the Project Site

12.5.33 No built heritage, including declared monuments, proposed monuments, graded historic buildings or government historic sites identified by the AMO, is identified within the Project Site. Therefore, no direct impact to built heritage is anticipated.

12.5.34 Nine other identified items (i.e. neither declared monument (site of cultural heritage) nor graded historic building), of which preservation, conservation and protection is not required under Annexes 10 and 19 of EIAO-TM, located within the Project Site are listed below:

i. Wai Cheung Ancestral Hall (HB1219);
ii. DD104 Lot 4186 S.E (Residence) (YTMT01);
iii. DD104 Lot 4187 S.B (Watchtower) (YTMT02);
iv. Remnants of Nam Shan Monastery (YTMT03);
v. Subsidiary Station of San Yau Vegetable Marketing Co-operative Society Ltd. (YTMT04);
vi. Former Yau Tam Mei Primary School (YTMT06);
vii. Lee's Boundary Stone (YTMT07);
viii. Mailbox No. 299 (NB04); and
ix. Grave of Mr. Man Chiu Pak and His Wife (NB09).

12.5.35 Wai Cheung Ancestral Hall (HB1219) is situated on land use of Open Space (O.3) in the RODP. In response to comments obtained during public engagement, with a view on its significance in cultural heritage, it will be preserved in situ. Therefore, potential indirect impacts of ground borne vibration, tilting and settlement and dust nuisance are anticipated on it during the construction phase.

12.5.36 Former Yau Tam Mei Primary School (YTMT06) is situated on land use of Government (UniTown) (G.11) in the RODP. As mentioned in **Section 2**, flexibility for potential adaptive reuse of the school structures by future project proponents of the UniTown for education / supporting facilities will be allowed in the RODP. Subject to the detailed design in later stages, the school may be preserved or demolished. If the school would be preserved in situ, potential indirect impacts of ground borne vibration, tilting and settlement and dust nuisance are anticipated on it during the construction phase. Otherwise, direct impact of demolition is anticipated on the school. While the actual impact of the proposed development to former Yau Tam Mei Primary School would be subject to further review in the next stage, this assessment would consider both the preserved and demolished scenarios for this item.

12.5.37 For the remaining seven other identified items located within the Project Site, they are situated on land uses of Government, Institution or Community (G.5 and G.8), Government (UniTown) (G.12), Private Residential (R.4), Road D1 and other road works in the RODP (**Figure 12.8** refers). Direct impact of demolition is therefore anticipated on these items.

Built Heritage and Other Identified Items within the 300 m Assessment Area but outside the Project Site

12.5.38 A total of forty-five built heritage and other identified items are located within the 300 m assessment area but outside the Project Site, including one grade 2 historic building, nine grade 3 historic buildings, and thirty-five other identified items.

12.5.39 Eight other identified items are located in proximity to the Project Site. Potential indirect impact of ground-borne vibration, tilting and settlement is anticipated on seven of them during the construction phase. For the remaining one other identified item, namely Man's Boundary Stone (NB05), no direct or indirect impact is anticipated for considering its physical nature. The seven other identified items in concern include:

i. San Yau Vegetable Marketing Co-operative Society, Ltd. (YTMT05);
ii. DD104 Lot 2729 (Residence) (WTT01);
iii. Nos. 16-17, San Wai Tsuen (SW03);

iv. No. 25A, San Wai Tsuen (SW04);
v. Mailbox No. 35 (NB02);
vi. Mailbox No. 169 (NB03); and
vii. Chun Chi Education Park (NB08).

12.5.40 San Yau Vegetable Marketing Co-operative Society, Ltd. (YTMT05) is located approximately 6 m from the Project Site. Potential indirect impact of dust nuisance is also anticipated on YTMT05 during the construction phase.

Operational Phase

12.5.41 No adverse impact on built heritage and other identified items is anticipated during operational phase, except Wai Cheung Ancestral Hall (HB1219) and former Yau Tam Mei Primary School (YTMT06) (if preserved in situ) at which direct impact is anticipated should they be utilised.

Mitigation Measures

Construction Phase

Cartographic and Photographic Record

12.5.42 Seven other identified items located within the Project Site, including YTMT01, YTMT02, YTMT03, YTMT04, YTMT07, NB04 and NB09, are proposed to be preserved by record if direct impact is imminent. Cartographic and photographic record, and other documentation means (including 3D scanning) should be conducted prior to the commencement of any construction works by the contractor(s) at the respective locations for record purposes and future use. For NB09, implementation details would be subject to discussion between project proponent(s) and stakeholders.

12.5.43 If demolition of YTMT06 is confirmed in subsequent stages, preserved by record should also be conducted for the school. Details should refer to **Section 12.5.42**.

Monitoring of Ground-borne Vibration, Tilting and Ground Settlement

12.5.44 Located within or adjacent to the Project Site, eight other identified items, including HB1219, YTMT05, WTT01, SW03, SW04, NB02, NB03 and NB08, may experience indirect impacts of ground-borne vibration, tilting and settlement. The future contractor(s) should implement standard control measures on ground-borne vibration, tilting and settlement by drawing necessary references from relevant government guidelines, including but not limited to the Code of Practice for Foundations[38] and Practice Note for Authorised Person, Registered Structural Engineers and Registered Geotechnical Engineers APP-137 (PNAP APP-137)[39].

12.5.45 During the construction phase, monitoring of ground-borne vibration, tilting and ground settlement should be implemented by the contractor(s) on these other identified items when there are construction works nearby. The monitoring should be incorporated with a set of Alert, Alarm and Action (3As) system. The proposed 3As criteria should be further confirmed via pre-construction condition survey and

[38] Buildings Department. (2024). *Code of Practice for Foundations 2017 (2024 Edition)*. Retrieved from https://www.bd.gov.hk/doc/en/resources/codes-and-references/code-and-design-manuals/FoundationCode2017.pdf.
[39] Buildings Department. (2024). *Practice Note for Authorised Person, Registered Structural Engineers and Registered Geotechnical Engineers (PNAP) – Ground-borne Vibration and Ground Settlement arising from Pile Foundation and Excavation and Lateral Support Work (APP-137)*. Retrieved from https://www.bd.gov.hk/doc/en/resources/codes-and-references/practice-notes-and-circular-letters/pnap/APP/APP137.pdf.

an assessment on the effects of ground-borne vibrations, settlements and tilting on the concerned buildings.

12.5.46 Considering that HB1219 is proposed to be preserved in situ within the Project Site, standard measures of pre-construction condition survey, as suggested in PNAP APP-137, should be conducted on HB1219 for better understanding on its structural condition. The survey should be undertaken by registered structural engineers or heritage specialists. The methodology for the condition survey should be proposed by the registered structural engineer(s) or heritage specialist(s). The results of the pre-construction condition survey should form a baseline and taken into consideration when formulating the abovementioned monitoring proposal (**Section 12.5.45** refers) and buffer zone (**Section 12.5.48** refers). A post-construction condition survey should also be carried out to confirm its structural stability.

12.5.47 If preservation of YTMT06 in situ is confirmed in subsequent stages, the abovementioned standard control measures (i.e. condition survey, ground-borne vibration, tilting and ground settlement) should also be conducted for the school. Details should refer to **Sections 12.5.44** to **12.5.46**.

Buffer Zone

12.5.48 A buffer zone should be reserved for HB1219 by the project proponent or its contractor(s) according to the results of the pre-construction condition survey (**Section 12.5.46** refers) in the design layout of the Project for mitigating potential adverse vibration impact arising from construction works. No piling works should be allowed within the buffer zone during the construction phase. The said buffer zone could be contained within the proposed Open Space O.3.

12.5.49 If preservation of YTMT06 in situ is confirmed in subsequent stages, same measures should also be conducted for the school. Details should refer to **Section 12.5.48**.

Physical Barriers

12.5.50 The contractor(s) should enforce protocol to forbid any light machinery, such as handheld jackhammer, or heavy machinery to come into direct contact with HB1219. Protective covering or sheltering should be provided for HB1219 during construction activities in proximity to avoid potential damages through direct contact with construction machineries. Physical protective barriers/covers, intervention/cushioning materials, or structural strengthening with minimal impact to the building fabric might need to be set up to protect the building if necessary.

12.5.51 If preservation of YTMT06 in situ is confirmed in subsequent stages, physical barriers should be also deployed for the school during construction phase. Details should refer to **Section 12.5.50**.

Dust Suppression

12.5.52 As HB1219, YTMT05 and YTMT06 (if preserved in situ) are located within or in close proximity of the Project Site, dust nuisance from the construction works nearby might have potential impact to the buildings and their users.

12.5.53 *Air Pollution Control (Construction Dust) Regulation* shall be followed. Dust suppression measures and good site practice should be adopted by the contractor(s) during the construction phase in order to avoid dust nuisance on HB1219, YTMT05 and YTMT06 (if preserved in situ).

Temporary Change of Access

12.5.54 There would be a temporary change of access to HB1219, YTMT05 and YTMT06 (if preserved in situ) during the construction phase. A safe access route to the concerned buildings should be maintained by the contractor(s) for conducting any mitigation measures.

Awareness to Other Identified Items in Close Proximity

12.5.55 A total of eight other identified items are located in proximity to the Project Site, namely YTMT05, WTT01, SW03, SW04, NB02, NB03, NB05 and NB08. To ensure no direct disturbance would be caused to the physical fabrics of these items, project proponent(s), subsequent developer(s) and contractor(s) should be aware of these items when construction works are carrying out nearby. Management measures should be deployed by the contractor(s), such as briefing to site staff before commencement of construction works and posting notices at site office(s).

Operational Phase

Revitalisation

12.5.56 HB1219 is proposed to be preserved in situ within the land use of Open Space (O.3) (**Figure 12.8** refers). Any revitalisation proposed for the building in later stages should be further reviewed by the future project proponent(s) or subsequent developer(s).

12.5.57 YTMT06 is located within the land use of Government (UniTown) (G.11) (**Figure 12.8** refers). If the preservation of YTMT06 in situ is confirmed in subsequent stages, opportunity for potential revitalisation of the school should be explored by future project proponent(s) or subsequent developer(s).

12.5.58 In view of no impact on built heritage and other identified items except the preserved in situ HB1219 and the possible preserved in situ YTMT06 during the operational phase, mitigation measure is not required.

12.5.59 A summary on the potential impact and the proposed mitigation measures for other identified items is presented in **Table 12.2**.

Environmental Monitoring and Audit

12.5.60 The implementation of the recommended mitigation measures mentioned in **Sections 12.5.42** to **12.5.58** should be audited as part of the EM&A programme. Details of the EM&A requirements are provided in a stand-alone EM&A Manual.

Conclusion

12.5.61 Baseline study, comprising of desktop research and field evaluation, has identified ten graded historic buildings located within assessment area but outside the Project Site. No built heritage is located within the Project Site. No direct or indirect impact is anticipated on the built heritage within the assessment area.

Construction Phase

12.5.62 A total of seven other identified items (i.e. DD104 Lot 4186 S.E (Residence) (YTMT01); DD104 Lot 4187 S.B (Watchtower) (YTMT02); Remnants of Nam Shan Monastery (YTMT03); Subsidiary Station of San Yau Vegetable Marketing Co-operative Society, Ltd. (YTMT04); Lee's Boundary Stone (YTMT07); Mailbox No. 299 (NB04); and the Grave of Mr. Man Chiu Pak and His Wife (NB09)) located

within the Project Site will be subject to direct impact due to demolition. Cartographic and photographic record, and other documentation means (including 3D scanning), should be carried out for these buildings/structures prior to the commencement of the construction works at the respective locations for record purposes and future use. For NB09, implementation details would be subject to discussion between project proponent(s) and stakeholders. If former Yau Tam Mei Primary School (YTMT06) is confirmed to be demolished in subsequent stages, preservation by record should also be conducted for the school.

12.5.63 Potential indirect impacts of ground-borne vibration, settlement and tilting is anticipated during the construction phase for eight other identified items (i.e. Wai Cheung Ancestral Hall (HB1219), San Yau Vegetable Marketing Co-operative Society, Ltd. (YTMT05); DD104 Lot 2729 (Residence) (WTT01); Nos. 16-17, San Wai Tsuen (SW03); No. 25A, San Wai Tsuen (SW04); Mailbox No. 35 (NB02); Mailbox No. 169 (NB03) and Chun Chi Education Park (NB08)) which are located within or in proximity to the Project Site. Standard control measures on ground-borne vibration, tilting and settlement shall be implemented by the future contractor(s) by drawing necessary references from relevant government guidelines.

12.5.64 A total of eight other identified items are located in proximity of the Project Site, namely YTMT05, WTT01, SW03, SW04, NB02, NB03, Man's Boundary Stone (NB05) and NB08. To ensure no direct disturbance would be caused to the physical fabrics of these items, project proponent(s), subsequent developer(s) and contractor(s) should be aware of these items when the construction works are carrying out nearby. Management measures should be deployed, such as briefing to site staff before commencement of construction works and posting notices at site office(s).

12.5.65 No impact is anticipated for the remaining built heritage and other identified items during the construction phase. Therefore, no mitigation measure is required.

Operational Phase

12.5.66 For operational phase, HB1219 will be preserved in situ, while YTMT06 would be either preserved in situ or demolished subject to detailed design stage. It is anticipated these other identified items would be utilised as part of their proposed land use, direct impact would be anticipated.

12.5.67 In view of no impact on built heritage and other identified items except the preserved in situ HB1219 and the possible preserved in situ YTMT06 during the operational phase, mitigation measure is not required.

Table 12.2 Summary of Potential Impact and the Proposed Mitigation Measures on Other Identified Items

Ref. No.	Name	Status	Approx. Distance to Project Site	Potential Impact		Mitigation Measures
				Construction Phase	Operational Phase	
Yau Tam Mei Tsuen						
HB1219	Wai Cheung Ancestral Hall	AAB discussed in 2010 and confirmed as no grading	Within Project Site	Potential indirect impacts of ground-borne vibration, tilting and ground settlement and dust nuisance are anticipated due to construction works in proximity.	If it will be preserved in situ and would be utilised as part of its proposed land use, direct impact would be anticipated.	Construction Phase (i) Standard control measures on ground-borne vibration, tilting and settlement should be conducted by the contractor(s), including condition surveys both before and after all construction works and monitoring of ground-borne vibration, tilting and ground settlement when there are construction works nearby. (ii) A buffer zone should be reserved by the project proponent or its contractor(s) according to the results of condition survey in the design layout of the Project. No piling works should be allowed within the buffer zone during the construction phase. (iii) Physical barriers should be provided by the contractor(s) during construction activities in proximity to avoid potential damages through direct contact with construction machineries. (iv) *Air Pollution Control (Construction Dust) Regulation* shall be followed. Dust suppression measures and good site practice should be adopted by the contractor(s). (v) A safe access route should be maintained by the contractor(s) for conducting mitigation measures during the construction phase.

Ref. No.	Name	Status	Approx. Distance to Project Site	Potential Impact		Mitigation Measures
				Construction Phase	Operational Phase	
						Operational Phase (vi) Any revitalisation proposed for the building in subsequent stages should be further reviewed by the future project proponent(s) or subsequent developer(s).
YTMT01	DD104 Lot 4186 S.E (Residence)	No Status	Within Project Site	Direct impact is anticipated.	No impact	Construction Phase Preservation by record via cartographic and photographic record, and other documentation means (including 3D scanning), should be conducted by the contractor(s) prior to the commencement of construction works at the respective location.
YTMT02	DD104 Lot 4187 S.B (Watchtower)	No Status	Within Project Site	Direct impact is anticipated.	No impact	Construction Phase Preservation by record via cartographic and photographic record, and other documentation means (including 3D scanning), should be conducted by the contractor(s) prior to the commencement of construction works at the respective location.
YTMT03	Remnants of Nam Shan Monastery	No Status	Within Project Site	Direct impact is anticipated.	No impact	Construction Phase Preservation by record via cartographic and photographic record, and other documentation means (including 3D scanning), should be conducted by the contractor(s) prior to the commencement of construction works at the respective location.

Ref. No.	Name	Status	Approx. Distance to Project Site	Potential Impact		Mitigation Measures
				Construction Phase	Operational Phase	
YTMT04	Subsidiary Station of San Yau Vegetable Marketing Co-operative Society, Ltd.	No Status	Within Project Site	Direct impact is anticipated.	No impact	Construction Phase Preservation by record via cartographic and photographic record, and other documentation means (including 3D scanning), should be conducted by the contractor(s) prior to the commencement of construction works at the respective location.
YTMT05	San Yau Vegetable Marketing Co-operative Society, Ltd.	No Status	6 m	Potential indirect impacts of ground-borne vibration, tilting and ground settlement and dust nuisance are anticipated due to construction works in proximity.	No impact	Construction Phase (i) Standard control measures on ground-borne vibration, tilting and settlement should be conducted by the contractor(s), including monitoring of ground-borne vibration, tilting and ground settlement when there are construction works nearby. (ii) *Air Pollution Control (Construction Dust) Regulation* shall be followed. Dust suppression measures and good site practice should be adopted by the contractor(s). (iii) A safe access route should be maintained by the contractor(s) for conducting mitigation measures during the construction phase. (iv) To ensure no direct disturbance would be caused to its physical fabrics, management measures should be deployed by the contractor(s), such as briefing to site staff before commencement of

Ref. No.	Name	Status	Approx. Distance to Project Site	Potential Impact		Mitigation Measures
				Construction Phase	Operational Phase	
						construction works and posting notices at site office(s).
YTMT06	Former Yau Tam Mei Primary School	No Status	Within Project Site	Subject to the detailed design in subsequent stages, if the school would be preserved in situ, potential indirect impacts of ground-borne vibration, tilting and ground settlement and dust nuisance are anticipated due to construction works in proximity.	If it will be preserved in situ and would be utilised as part of its proposed land use, direct impact would be anticipated.	<u>Construction Phase</u> (i) Standard control measures on ground-borne vibration, tilting and settlement should be conducted by the contractor(s), including condition surveys both before and after all construction works and monitoring of ground-borne vibration, tilting and ground settlement when there are construction works nearby. (ii) A buffer zone should be reserved by the project proponent or its contractor(s) according to the results of condition survey in the design layout of the Project. No piling works should be allowed within the buffer zone during the construction phase. (iii) Physical barriers should be provided by the contractor(s) during construction activities in proximity to avoid potential damages through direct contact with construction machineries. (iv) *Air Pollution Control (Construction Dust) Regulation* shall be followed. Dust suppression measures and good site practice should be adopted by the contractor(s). (v) A safe access route should be maintained by the contractor(s) conducting mitigation measures during the construction phase.

Ref. No.	Name	Status	Approx. Distance to Project Site	Potential Impact		Mitigation Measures
				Construction Phase	Operational Phase	
						Operational Phase (vi) Opportunity for potential revitalisation should be explored and mitigation measures will be proposed as appropriate by the future project proponent(s) or subsequent developer(s) in the detailed design stage.
				Subject to the detailed design in subsequent stages, if the school would not be preserved, direct impact is anticipated.	No impact	Construction Phase Preservation by record via cartographic and photographic record, and other documentation means (including 3D scanning), should be conducted by the contractor(s) prior to the commencement of construction works at the respective location.
YTMT07	Lee's Boundary Stone	No Status	Within Project Site	Direct impact is anticipated.	No impact	Construction Phase Preservation by record via cartographic and photographic record, and other documentation means (including 3D scanning), should be conducted by the contractor(s) prior to the commencement of construction works at the respective location.
Wai Tsai Tsuen						
WTT01	DD104 Lot 2729 (Residence)	No Status	25 m	Potential indirect impacts of ground-borne vibration, tilting and ground settlement are anticipated due to	No impact	Construction Phase (i) Standard control measures on ground-borne vibration, tilting and settlement should be conducted by the contractor(s), including monitoring of ground-borne vibration, tilting and

Ref. No.	Name	Status	Approx. Distance to Project Site	Potential Impact		Mitigation Measures
				Construction Phase	Operational Phase	
				construction works in proximity.		ground settlement when there are construction works nearby. (ii) To ensure no direct disturbance would be caused to its physical fabrics, management measures should be deployed by the contractor(s), such as briefing to site staff before commencement of construction works and posting notices at site office(s).
San Wai						
SW03	Nos. 16-17, San Wai Tsuen	No Status	48 m	Potential indirect impacts of ground-borne vibration, tilting and ground settlement are anticipated due to construction works in proximity.	No impact	<u>Construction Phase</u> (i) Standard control measures on ground-borne vibration, tilting and settlement should be conducted by the contractor(s), including monitoring of ground-borne vibration, tilting and ground settlement when there are construction works nearby. (ii) To ensure no direct disturbance would be caused to its physical fabrics, management measures should be deployed by the contractor(s), such as briefing to site staff before commencement of construction works and posting notices at site office(s).
SW04	No. 25A, San Wai Tsuen	No Status	20 m	Potential indirect impacts of ground-borne vibration, tilting and ground settlement are anticipated due to	No impact	<u>Construction Phase</u> (i) Standard control measures on ground-borne vibration, tilting and settlement should be conducted by the contractor(s), including monitoring of ground-borne vibration, tilting and

Ref. No.	Name	Status	Approx. Distance to Project Site	Potential Impact		Mitigation Measures
				Construction Phase	Operational Phase	
				construction works in proximity.		ground settlement when there are construction works nearby. (ii) To ensure no direct disturbance would be caused to its physical fabrics, management measures should be deployed by the contractor(s), such as briefing to site staff before commencement of construction works and posting notices at site office(s).
General (NTM Area)						
NB02	Mailbox No. 35	No Status	13 m	Potential indirect impacts of ground-borne vibration, tilting and ground settlement are anticipated due to construction works in proximity.	No impact	<u>Construction Phase</u> (i) Standard control measures on ground-borne vibration, tilting and settlement should be conducted by the contractor(s), including monitoring of ground-borne vibration, tilting and ground settlement when there are construction works nearby. (ii) To ensure no direct disturbance would be caused to its physical fabrics, management measures should be deployed by the contractor(s), such as briefing to site staff before commencement of construction works and posting notices at site office(s).
NB03	Mailbox No. 169	No Status	37 m	Potential indirect impacts of ground-borne vibration, tilting and ground settlement are anticipated due to	No impact	<u>Construction Phase</u> (i) Standard control measures on ground-borne vibration, tilting and settlement should be conducted by the contractor(s), including monitoring of ground-borne vibration, tilting and

Ref. No.	Name	Status	Approx. Distance to Project Site	Potential Impact		Mitigation Measures
				Construction Phase	Operational Phase	
				construction works in proximity.		ground settlement when there are construction works nearby. (ii) To ensure no direct disturbance would be caused to its physical fabrics, management measures should be deployed by the contractor(s), such as briefing to site staff before commencement of construction works and posting notices at site office(s).
NB04	Mailbox No. 299	No Status	Within Project Site	Direct impact is anticipated.	No impact	<u>Construction Phase</u> Preservation by record via cartographic and photographic record, and other documentation means (including 3D scanning), should be conducted by the contractor(s) prior to the commencement of construction works at the respective location.
NB05	Man's Boundary Stone	No Status	40 m	No impact	No impact	<u>Construction Phase</u> To ensure no direct disturbance would be caused to its physical fabrics, management measures should be deployed by the contractor(s), such as briefing to site staff before commencement of construction works and posting notices at site office(s).
NB08	Chun Chi Education Park	No Status	33 m	Potential indirect impacts of ground-borne vibration, tilting and ground settlement are anticipated due to	No impact	<u>Construction Phase</u> (i) Standard control measures on ground-borne vibration, tilting and settlement should be conducted by the contractor(s), including monitoring of ground-borne vibration, tilting and

Ref. No.	Name	Status	Approx. Distance to Project Site	Potential Impact		Mitigation Measures
				Construction Phase	Operational Phase	
				construction works in proximity.		ground settlement when there are construction works nearby. (ii) To ensure no direct disturbance would be caused to its physical fabrics, management measures should be deployed by the contractor(s), such as briefing to site staff before commencement of construction works and posting notices at site office(s).
NB09	Grave of Mr. Man Chiu Pak and His Wife	No Status	Within Project Site	Direct impact is anticipated.	No impact	<u>Construction Phase</u> Preservation by record via cartographic and photographic record, and other documentation means (including 3D scanning), should be conducted by the contractor(s) prior to the commencement of construction works at the respective location. Implementation details would be subject to discussion between project proponent(s) and stakeholders.

12.6 Archaeological Impact Assessment

Baseline Condition

12.6.1 Archaeological heritage are the places/areas of archaeological significance, or where the archaeological potential is considered noteworthy. The archaeological heritage exclusively includes the following:

- SAI in Hong Kong: under the *List of Sites of Archaeological Interest in Hong Kong* maintained by the AMO; and
- Area of archaeological potential identified in previous studies and / or this Project, but not included in the item above.

Desktop Study

Past Archaeological Investigations and Assessments

12.6.2 Over 70% (approximately 346 hectares) of the assessment area has been archaeologically studied in the past. One SAI, namely Ngau Tam Mei SAI (SAI38), is located partially within the Project Site. During the Territorial-Wide Archaeological Survey conducted in 1997, artefacts dated to the Bronze Age and the *Han* dynasty were discovered (**Section 12.4.30** and **Figure 12.5** refer) on this location. According to the findings of further archaeological survey conducted in 1999 for *The Main Drainage Channels for Ngau Tam Mei Project, Phase II*, twenty-two auger holes and one test trench were conducted within the nullah project boundary, and twelve auger holes and three trenches were carried out in the vicinity of the Ngau Tam Mei SAI. Only pottery sherds dated to the late Bronze Age and tiles dated to the *Han* dynasty were found on the surface of the SAI. Thus, its archaeological potential is determined unlikely to extend beyond the SAI boundary (**Sections 12.4.31** to **12.4.32** and **Figure 12.5** refer).

12.6.3 Archaeological survey was also conducted in 2023 presented in the NOL Main Line EIA report (**Sections 12.4.34** to **12.4.36** and **Figure 12.5** refer). Three test pits along with field scanning were carried out in NTM area, which are located within the Project Site, but neither archaeological features nor artefacts were discovered.

12.6.4 For the AIA conducted under STLMC DN EIA report, about 52 hectares of land in San Tin were field-scanned, and two test pits were excavated near Chau Tau Tsuen. In particular, field scanning area A9 located near Shek Wu Wai San Tsuen has encroached into the assessment area of this Project, but outside the Project Site. As the area has been significantly disturbed and no archaeological material was identified, it was considered to have no archaeological potential (**Sections 12.4.37** to **12.4.38** and **Figure 12.5** refer).

12.6.5 In addition, the archaeological potential of the majority of the Project Site and the assessment area had also been identified in previous archaeological assessments, including the EIA studies for *Stage 2 of PWP Item No. 215DS – Yuen Long and Kam Tin Sewerage and Sewage*, *NOL Main Line*, *STLMC DN*, *and NTMWTW* (**Sections 12.4.33** to **12.4.39** and **Figures 12.6.1** to **12.6.4** refer).

Comparison with Archaeological Sites in Similar Landscape Settings

12.6.6 For the remaining areas with no previous archaeological investigations or assessment conducted before, its archaeological potential could be deduced by understanding its environment and the settlement patterns in the NTM area.

12.6.7 Previous archaeological findings in Hong Kong suggested past human occupation on relatively flat area along coastal beaches as attractive places for settlement during the prehistoric period[40]. Considering the geographical and geological characteristics, the NTM area primarily encompasses low-lying areas that consist of plains, river valleys and fishponds. The area extends towards the foothills, with the hill ranges of *Kai Kung Leng* located to the south and Deep Bay situated to the west (**Sections 12.4.2** to **12.4.4** refer). The low-lying plains close to the Deep Bay at the west of NTM area might be favourable to past humans.

12.6.8 For historical period, traditional settlements such as Wai Tsai Tsuen and Chuk Yuen Tsuen can be dated back to the *Qing* dynasty (**Sections 12.4.21** to **12.4.24**, and **Figure 12.3** refer). They are examples of human settlements near the original coast of Deep Bay. Based on these existing references, it is likely that traditional villages in NTM area if once existed were predominantly situated on the estuary or a sheltered landscape at relatively higher elevations. The presence of Wai Tsai Tsuen and Chuk Yuen Tsuen in the west of the NTM valley indicates favourable landscapes for human settlement during the historical period.

Disturbance to Archaeological Remains

12.6.9 It is also important to acknowledge that certain areas within the assessment area may have experienced disturbances, such as changes in land use and alterations to the landscape. These disturbances may have impacted the preservation of archaeological information in those specific areas. As a result, it is possible to identify areas where the soil had been disturbed and where no archaeological potential is expected to be present. These locations include:

- Former offshore area;
- Past and current watercourses and water bodies, such as natural streams and ponds; and
- Disturbed or developed areas, such as cut slopes and areas of development(s) where site formation works had been carried out.

Evaluation of Archaeological Potential

12.6.10 Archaeological potential refers to the possibility of finding archaeological remains, artefacts, or sites in a particular area. It is an assessment of the chances of discovering valuable historical or cultural information through archaeological investigation, which the purpose is to uncover, study and interpret archaeological information to gain insights into past humans. Based on the above evaluation, the archaeological potential within the assessment area has been identified and are grouped into the following categories:

- *No Archaeological Potential Area* – Where the existence of archaeological remains is deemed improbable. This includes former coastal areas unsuitable for human settlement due to past submersion, steep slopes unfavourable for archaeological deposition due to gravity, significantly disturbed or developed sites where any potential archaeological deposits have been eradicated, and areas previously confirmed to be lack of archaeological potential by previous archaeological surveys or investigations.

[40] 商志醰 、吳偉鴻 (2010) 。《香港考古學叙研》。北京：文物出版社。

- *Low Archaeological Potential Area* – Areas previously confirmed to have low archaeological potential by previous archaeological surveys or investigations fall under this classification.

- *Moderate-Low Archaeological Potential Area* – Where archaeological remains may have existed in the past, but their preservation has been compromised by historical disturbances, modern development activities, or geological characteristics. Locations susceptible to flooding or with unfavourable settlement conditions are also classified as moderate-low archaeological potential areas.

- *High Archaeological Potential Area* – Where archaeological sites, finds spots, and/or standing structures, or geological features suggest past human settlement. These areas have experienced minimal disturbance or development, are identified within SAIs by the AMO, or have been confirmed to possess significant archaeological potential through prior surveys or investigations.

12.6.11 The archaeological potential within the 300 m assessment area is shown in **Figure 12.10**. Details of discussion on the archaeological potential within the Project Site are presented in the following sections.

12.6.12 To facilitate the discussion in the next sections on impact assessment and mitigation measures, a specific term "Archaeologically Sensitive Area (ASA)" is adopted to refer the area having high archaeological potential within the Project Site that require to be further surveyed/studied.

No Archaeological Potential Area

- *Past Archaeological Investigations and Assessments (**Figure 12.9** refers)*

Archaeological Investigation Report on The Main Drainage Channels for Ngau Tam Mei Project, Phase II[41]

12.6.13 An archaeological survey was conducted in 1999 for the construction of channelised nullah in NTM (**Sections 12.4.31** to **12.4.32** refer). Twenty-two auger holes and one trench were dug within and in close proximity to the study area of the nullah project (Plate 1 in **Appendix 12.2** and **Figure 12.5** refer). Neither archaeological features nor artefacts were discovered. Therefore, it can be concluded that there is no archaeological potential within the study area of the nullah project.

EIA and Traffic Impact Assessment (TIA) Studies for the Stage 2 of PWP Item No. 215DS – Yuen Long and Kam Tin Sewerage and Sewage Disposal – Environmental Impact Assessment Report (Register No.: AEIAR-078/2004) [42]

12.6.14 An EIA study was conducted in 2004 for the Yuen Long and Kam Tin Sewerage and Sewage Disposal project (**Section 12.4.33** refers). The EIA study covered the provision of sewer from Kam Tin River to NTM, Tam Mei Camp and San Tin Camp within the assessment area. The archaeological potential of these areas was examined in the EIA study, and the findings indicated that these areas do not possess any archaeological potential as these areas were found to be characterised by former ponds, estuarine sediments, river meandering, and roads,

[41] 古物古蹟辦事處 (1999) 。《元朗牛潭尾主渠道第二期工程考古調查工作報告》。香港：古物古蹟辦事處檔案。
[42] Drainage Services Department. (2013). *EIA and TIA Studies for the Stage 2 of PWP Item No. 215DS – Yuen Long and Kan Tin Sewerage and Sewage Disposal (Agreement No. CE 66/2001). Section 15 Cultural Heritage Assessment.* Retrieved from https://www.epd.gov.hk/eia/files/applications/en/pp_82/eia_1578/progress/action_3229/eia_report.pdf.

which had subject to cumulative impacts from various utilities. Thus, there is no archaeological potential as the area was unsuitable for human settlement due to past submersion and any potential archaeological deposits would have been eradicated.

Environmental Impact Assessment Study for Northern Link – Environmental Impact Assessment Report (Register No.: AEIAR-259/2024)[43]

12.6.15 An archaeological survey was conducted in 2023 presented in the NOL Main Line EIA report. NOL Main Line connects Kam Sheung Road (錦上路) of Yuen Long District and Kwu Tung (古洞) of North District. Three test pits along with field scanning were carried out in NTM area, which are located within the Project Site. In addition, field scanning was conducted at Long Ha Tsuen, which is located within the 300 m assessment area but falls outside the Project Site (**Sections 12.4.34** to **12.4.36** refer). Neither archaeological features nor artefacts were discovered in the surveyed area in NTM area. Therefore, no archaeological potential can be concluded for these surveyed areas.

First Phase Development of the New Territories North – San Tin / Lok Ma Chau Development Node – Environmental Impact Assessment Report (Register No.: AEIAR-261/2024) [44]

12.6.16 Under the STLMC DN EIA report, field scanning at area A9 was conducted near Shek Wu Wai San Tsuen, which encroaches into the assessment area of this Project, but is outside the Project Site (**Sections 12.4.37** to **12.4.38** refer). The field scanning concluded that the scanned area contains many modern trashes and signs of site flattening in recent times that would have already disturbed the ground with no archaeological material identified. Thus, the area has no archaeological potential due to the area being significantly disturbed.

Ngau Tam Mei Water Treatment Works Extension – Environmental Impact Assessment Report (Register No.: AEIAR-262/2024)[45]

12.6.17 A desktop review was conducted for the EIA study on the extension of Ngau Tam Mei Water Treatment Works. The proposed works do not encroach into the Project Site, but fall within the 300 m assessment area (**Section 12.4.39** refers). Based on the evaluation of the area's geological conditions, the proposed works are either located on a hilly landscape that is unfavourable for human settlement, or on the existing road network, disturbed areas with existing utilities and cut slope. As such, these areas are considered to have no archaeological potential due to steep slope or significant disturbance.

- *Review of Past Fieldwork Data*

12.6.18 Previous fieldworks have been conducted within the 300 m assessment area and could infer further on the archaeological potential of the area (**Figure 12.5** refers).

[43] MTR Corporation Limited. (2023). *Environmental Impact Assessment Study for Northern Link (Agreement No. C1603). Section 13 Cultural Heritage Impact Assessment.* Retrieved from https://www.epd.gov.hk/eia/files/applications/en/pp_346/eia_1992/progress/action_1365/eia_3012023/HTML/EIA%20Report/S13%20CHIA.htm.

[44] Civil Engineering and Development Department and Planning Department. (2023). *First Phase Development of the New Territories North – San Tin / Lok Ma Chau Development Node – Investigation (Agreement No. CE 20/2021). Section 12 Impact on Cultural Heritage.* Retrieved from https://www.epd.gov.hk/eia/files/applications/en/pp_340/eia_1994/progress/action_157674/eia_3022023/EIA/CE20_EIA_S12.htm#_Toc154070604.

[45] Water Supplies Department. (2024). *Ngau Tam Mei Water Treatment Works Extension – Environmental Impact Assessment Report (282551-REP-080-05). Section 11 Impact on Cultural Heritage.* Retrieved from https://www.epd.gov.hk/eia/register/study/latest/esb-333.pdf.

12.6.19 The 1999 archaeological investigation for *The Main Drainage Channels for Ngau Tam Mei Project, Phase II*[46] involved hand augering at locations A6 to A10, A12 and A13[47] for the main drainage area to the west of Yau Tam Mei village. All auger holes revealed a brownish-yellow sandy clay immediately beneath the topsoil.

12.6.20 Similarly, the 2023 archaeological works presented in the NOL Main Line EIA report[48] identified the same brownish-yellow sandy clay as sterile layer in context C3 of both test pits NTM-TP1 and NTM-TP2 in Ngau Tam Mei. This investigation concluded that this layer is composed of alluvium. The areas surrounding test pits NTM-TP1 and NTM-TP2 demonstrate low archaeological potential, due to the lack of archaeological materials and the relatively recent formation of land from alluvial deposition by the local river.

12.6.21 Given that both surveys found no archaeological materials, it is reasonable to conclude that the areas surrounding A6 to A10, A12 and A13 for the main drainage area have no archaeological potential, with a sterile layer present beneath the topsoil.

- *Former Offshore Area (**Figure 12.9** refers)*

12.6.22 The former offshore areas, as determined by geological deposits, indicate that they were previously submerged under water before modern development took place. These areas are characterised by superficial deposits of marine mud (*QHH* deposit on **Figure 12.2**) and estuarine deposits (*Qam* deposit on **Figure 12.2**) (**Section 12.4.4** refers)[49]. Due to their unsuitability for human settlement, areas beyond the former coastline which were once part of the sea are unlikely to possess significant archaeological potential due to past submersion.

- *Past and Current Watercourses and Water Bodies (**Figure 12.9** refers)*

12.6.23 There are natural rivers, channelised rivers and fishponds within the Project Site and assessment area in the past and present. Plates 2 to 8 in **Appendix 12.2** depicted the fishponds and watercourses that existed in the NTM valley as observed in aerial photos and topographic map. Due to past and current submersion, it is unlikely for these watercourses and water bodies to hold archaeological potential. Moreover, the construction of channelised rivers and ponds would render the location with no archaeological potential as the construction works would have eradicated any potential archaeological deposits.

12.6.24 Moreover, the meandering patterns observed to the north of the Ngau Tam Mei Drainage Channel indicate that the flow of water has consistently influenced the surrounding landscape, including the flood plain (*Qa* deposit on **Figure 12.2**) (**Section 12.4.4** refers)[50]. This process involves the erosion of soils and their deposition into the sea towards the Deep Bay. Consequently, the flood plain is unlikely to hold significant archaeological potential, as past submersion has not been conducive to the preservation of archaeological remains in their original positions.

[46] 古物古蹟辦事處 (1999)。《元朗牛潭尾主渠道第二期工程考古調查工作報告》。香港：古物古蹟辦事處檔案。
[47] A11 is located within the said drainage works area.
[48] MTR Corporation Limited. (2023). *Environmental Impact Assessment Study for Northern Link (Agreement No. C1603). Section 13 Cultural Heritage Impact Assessment.* Retrieved from https://www.epd.gov.hk/eia/files/applications/en/pp_346/eia_1992/progress/action_1365/eia_3012023/HTML/EIA%20Report/S13%20CHIA.htm.
[49] Geotechnical Control Office. (1989). *Hong Kong Geological Survey Sheet 2*. Government of Hong Kong.
[50] *Ibid*.

- *Disturbed or Developed Area (**Figure 12.9** refers)*

12.6.25 Desktop study has revealed that the Project Site has undergone extensive modern development, primarily driven by infrastructure projects such as the channelisation of NTM river, the NTMWTW and slope cutting. Additionally, there are also construction of public and private property developments in the NTM area, such as the Transitional Housing Project at Ngau Tam Mei South (Chun Shin Road)[51], ***Casa Paradizo*** and ***The Vineyard***. These modern construction activities had significantly disturbed the natural soil, as well as any archaeological deposits may have present within it. As a result, the developed or disturbed areas within the assessment area are considered to have no archaeological potential as any potential archaeological deposits have been eradicated in these developed sites**.**

*Low Archaeological Potential Area (**Figure 12.10** refers)*

- *Past Archaeological Investigations and Assessments*

Environmental Impact Assessment Study for Northern Link – Environmental Impact Assessment Report (Register No.: AEIAR-259/2024)[52]

12.6.26 The archaeological survey conducted in NTM area shows that there was no human settlement in or near the surveyed area before modern times. The fieldworks also revealed this area has a high underground water table. Thus, the surveyed NTM area in the NOL Main Line EIA report is concluded to have low archaeological potential (**Sections 12.4.34** to **12.4.36** refer).

12.6.27 In addition to the archaeological survey results, the archaeological prediction model had assessed the likelihood of archaeological deposits for areas that have not been surveyed or confirmed in previous archaeological assessments within the assessment area in the NOL Main Line EIA report.

12.6.28 The continuous agricultural activities that took place, coupled with persistent high water level, suggested that the surface soil of these agricultural fields had underwent significant disturbance. It rendered these areas inhospitable for lasting human habitation. Hilly terrain with steep slopes were also expected to be not suitable for human settlements. This resounding absence of any evidence reinforces the notion that sustained settlement was impractical in these areas in both historical and prehistoric period. Consequently, it can be concluded that these areas hold low archaeological potential and were unsuitable for past human habitation.

First Phase Development of the New Territories North – San Tin / Lok Ma Chau Development Node – Environmental Impact Assessment Report (Register No.: AEIAR-261/2024) [53]

12.6.29 The STLMC DN EIA report had identified areas with no considerable amount of archaeological discovery nor in close proximity to the existing traditional villages, are unlikely to contain archaeological potential. Shek Wu Wai San Tsuen located

[51] Housing Bureau. *Transitional Housing Project at Ngau Tam Mei South (Chun Shin Road), Yuen Long*. Retrieved from https://www.hb.gov.hk/eng/policy/housing/policy/transitionalhousing/details_70.html.
[52] MTR Corporation Limited. (2023). *Environmental Impact Assessment Study for Northern Link (Agreement No. C1603). Section 13 Cultural Heritage Impact Assessment.* Retrieved from https://www.epd.gov.hk/eia/files/applications/en/pp_346/eia_1992/progress/action_1365/eia_3012023/HTML/EIA%20Report/S13%20CHIA.htm.
[53] Civil Engineering and Development Department and Planning Department. (2023). *First Phase Development of the New Territories North – San Tin / Lok Ma Chau Development Node – Investigation (Agreement No. CE 20/2021). Section 12 Impact on Cultural Heritage.* Retrieved from https://www.epd.gov.hk/eia/files/applications/en/pp_340/eia_1994/progress/action_157674/eia_3022023/EIA/CE20_EIA_S12.htm#_Toc154070604.

near the field scanning area A9 was newly established in the 20th century and is therefore unlikely to contain any archaeological potential in historical period based on its settlement history.

12.6.30 The archaeological prediction model had also assessed the likelihood of archaeological deposits within the assessment area in the STLMC DN EIA report. The continuous agricultural activities that took place, coupled with persistent high water level, suggested that the surface soil underwent significant disturbance. The hill slopes along *Ngau Tam Shan* with higher grounds are inhospitable for lasting human habitation as well. Consequently, it can be concluded that these areas do not hold significant archaeological potential and considered as low archaeological potential area.

*Moderate-Low Archaeological Potential Area (**Figure 12.10** refers)*

- *Low-lying Agricultural Fields of the Upper Stream of NTM Area*

12.6.31 Archaeological investigations conducted prior to Ngau Tam Mei SAI have marked its archaeological potential likely located within its extent. As stated in earlier **Sections 12.4.30** to **12.4.32**, archaeological finds dating back to the Bronze Age and the early *Han* dynasty were only discovered in the vicinity of Ngau Tam Mei SAI. The latter *Archaeological Investigation Report on The Main Drainage Channels for Ngau Tam Mei Project, Phase II* [54] supplemented that its archaeological potential is unlikely to extend beyond the SAI boundary due to the absence of archaeological finds in the NTM river valley. However, the "unlikely to extend" is not clearly defined.

12.6.32 The NTM river valley has predominantly been utilised with agricultural fields and fishponds as observed from past topographic maps and aerial photographs[55]. While open storages and carparks occupied many of the NTM valley at the present, there are still some in situ agricultural fields and fishponds (or the remains of) since at least the 1960s and 1970s. Noting that the upper stream of the NTM river valley has an elevation from +8.6 mPD to the west and rising to +21.8 mPD in the east. Given such high elevations, fishponds are still distributed along the river valley up to +21.2 mPD[56] (Plate 9 in **Appendix 12.2** refers). The distribution of fishponds at such upstream, high elevations landscape suggested the presence of a consistently high underground water level, which is essential for the viability of these fishponds. On site observation suggested that the current fishponds maintained a stable water level at about 20cm below the ground surface.

[54] 古物古蹟辦事處 (1999)。《元朗牛潭尾主渠道第二期工程考古調查工作報告》。香港：古物古蹟辦事處檔案。
[55] Lands Department. (1966). *1:1,200. 75-SW-C [topographic map]*. Lands Department.
Lands Department. (1969). *1:1,200. 75-SW-C [topographic map].* Lands Department.
Lands Department. (1975). *1:1,200. 75-SW-C [topographic map].* Lands Department.
Lands Department. (1978). *1:1,200. 75-SW-C [topographic map].* Lands Department.
Lands Department. (1966). *1:1,200. 75-SW-D [topographic map].* Lands Department.
Lands Department. (1969). *1:1,200. 75-SW-D [topographic map].* Lands Department.
Lands Department. (1974). *1:1,200. 75-SW-D [topographic map].* Lands Department.
Lands Department. (1978). *1:1,200. 75-SW-D [topographic map].* Lands Department.
Lands Department. (1966). *1:1,200. 92-NW-A [topographic map].* Lands Department.
Lands Department. (1969). *1:1,200. 92-NW-A [topographic map].* Lands Department.
Lands Department. (1975). *1:1,200. 92-NW-A [topographic map].* Lands Department.
Lands Department. (1978). *1:1,200. 92-NW-A [topographic map].* Lands Department.
Lands Department. (1966). *1:1,200. 92-NW-B [topographic map].* Lands Department.
Lands Department. (1973). *1:1,200. 92-NW-B [topographic map].* Lands Department.
Lands Department. (1975). *1:1,200. 92-NW-B [topographic map].* Lands Department.
[56] Antiquities and Monuments Office. *Geographical Information System on Hong Kong Heritage*. Retrieved from https://gish.amo.gov.hk/internet/index.html?lang=en-us.

12.6.33 This finding is consistent with the findings from past archaeological investigations. Archaeological survey conducted in the NOL Main Line EIA report[57] revealed some underground water level information from a test pit *NTM-TP1*. This recorded a ground surface level of +7.2 mPD, with the underground water level observed at approximately +7.0 mPD (some 20 cm below the ground surface). Furthermore, under the same archaeological survey, two test pits *SAT-TP5* and *SAT-TP8* (Plate 10 in **Appendix 12.2** refers) were surveyed at San Tin river valley, where a similar landscape with many fish ponds exhibited the area since the 1960s. *SAT-TP5* recorded a ground surface level of +6.9 mPD and revealed underground water level at approximately +6.7 mPD, while *SAT-TP8* recorded a ground surface level of +7.6 mPD and revealed underground water at about +7.3 mPD.

12.6.34 The consistency of both field observations near fish ponds in the upper streams of NTM river valley and the past archaeological survey observations in similar landscapes at NTM and San Tin indicated the areas in proximity to fishponds are likely to have high underground water level.

12.6.35 Furthermore, the subtropical climate of Hong Kong means that the area would be subjected to considerable rainfall. About 2,000 mm of rainfall has been recorded annually [58] [59]. The heavy rainfall could easily surpass the capacity of the underground water storage of the soil in such a high underground water elevation and cause outflow on the ground surface. Hence, NTM river valley is susceptible to flooding. As a result, these areas are unlikely to be inhabited by past humans, as it is also reflected choice of location of existing settlements during the historical period in this area.

12.6.36 While the surrounding areas of Ngau Tam Mei SAI would seems to have lower archaeological potential based on the abovementioned environmental factors and past archaeological surveys results, these upper stream areas on the low-lying lands of NTM river valley is considered to have moderate-low archaeological potential due to its proximity to known archaeological finding spots of the SAI.

- *Hilly landscape of NTM Area*

12.6.37 The presence of hilly landscape to the sides of the NTM river valley with slopes gradient over 20° is unfavourable for settlement as historical settlements typically sought flatlands for habitation, prioritising sheltered environments that offered protection from various threats (**Section 12.4.4** refers). Furthermore, the hill slopes do not favour in situ archaeological deposits, and hence the archaeological potential is considered lower.

12.6.38 While the surrounding areas of Ngau Tam Mei SAI seem to have lower archaeological potential based on the abovementioned environmental factor, these hilly areas of NTM river valley are considered to have moderate-low archaeological potential due to its proximity to known archaeological finding spots of the SAI.

[57] MTR Corporation Limited. (2023). *Environmental Impact Assessment Study for Northern Link (Agreement No. C1603). Section 13 Cultural Heritage Impact Assessment.* Retrieved from https://www.epd.gov.hk/eia/files/applications/en/pp_346/eia_1992/progress/action_1365/eia_3012023/HTML/EIA%20Report/S13%20CHIA.htm.
[58] Hong Kong Observatory. (2025). *Yearly Extract. Climate Information Service*. Retrieved from https://www.hko.gov.hk/en/cis/yearlyExtract.htm.
[59] Fyfe, J. A et al. (2000). *The Quaternary Geology of Hong Kong*, p.26. Hong Kong: Civil Engineering Department.

*High Archaeological Potential Area (**Figure 12.10** refers)*

Ngau Tam Mei SAI

- *Previous Archaeological Surveys*

12.6.39 The Yuen Long terrestrial-wide archaeological survey conducted in 1997 yielded significant archaeological findings and identified the extent of Ngau Tam Mei SAI[60]. Pottery shards with distinctive "Union Jack" patterns dated back to the Bronze Age were discovered along with tiles dated to the *Han* dynasty. These findings indicate the presence of human settlements during prehistoric and historical periods in the area (**Section 12.4.30** refers).

12.6.40 Further investigation in the vicinity of Ngau Tam Mei SAI were conducted at the time of the archaeological investigation in 1999[61]. Although no archaeological findings were retrieved from the trench excavated, pottery sherds dated to the late Bronze Age and tiles from the *Han* dynasty were collected on the ground surface of the SAI. The presence of these significant archaeological findings confirms the area's importance in both prehistoric and historical periods and it contains high archaeological potential (**Sections 12.4.31** to **12.4.32** refer).

Ngau Tam Mei Area

- *Previous Archaeological Assessment*

12.6.41 Ngau Tam Mei Area is identified to have high archaeological potential in the NOL Main Line EIA report (**Section 12.4.36** refers). It is located in the west of the assessment area of this Project.

12.6.42 Ngau Tam Mei Area exhibits archaeological potential in prehistoric period. It is situated at the foot of a small mound within the river valley of NTM. The Ngau Tam Mei Area's elevation ranges from +8 mPD to +13 mPD and exhibits a gradient of 0° to 8°. Its environmental settings consist of river terraces near hillslopes, with geology predominantly featuring alluvial and colluvial deposits (*Qa*, *Qpa* and *Qpd*). These environmental characteristics resemble those of Sheung Shui Wah Shan SAI (SAI71), which is situated on a river terrace in front of a hillslope. It is comparable to the preferred landscapes for past human settlements in prehistoric period.

12.6.43 Ngau Tam Mei Area is also expected to have a high likelihood of archaeological deposits from the historical period. The area exhibiting archaeological potential in historical period is located at the mouth of a narrow river valley in NTM, facing the shallow sea of Deep Bay to the west, and positioned on the hillslope of Kai Kung Leng to the south. The elevation of this area ranges from +2 mPD to +24 mPD and the gradient varies from 0° to 18°. This river terrace and coastal landscape provide suitable conditions for human habitation during the historical period.

12.6.44 The portion falls within the Project Site with high archaeological potential is hereinafter referred to as Ngau Tam Mei Archaeological Sensitive Area (ASA).

12.6.45 The current assessment has reviewed the results of previous assessments and conducted a detailed review within the Project Site. Some areas have been further reviewed in earlier paragraphs and assessed to have no archaeological potential instead. Furthermore, NOL Main Line works area would be handed back to this Project once they have completed their construction works. Thus the construction

[60] 古物古蹟辦事處 (1997)。《全港文物普查 1997 第一地區(元朗區)工作報告》。香港：古物古蹟辦事處檔案。
[61] 古物古蹟辦事處 (1999)。《元朗牛潭尾主渠道第二期工程考古調查工作報告》。香港：古物古蹟辦事處檔案。

works of NOL Main Line would have exhausted the archaeological potential within its project area before handing over to this Project. Other than those areas previously discussed, this AIA agrees the archaeological potential assessment result of previous study over the remaining for Ngau Tam Mei Area.

Ngau Tam Mei (North) Area

- *Previous Archaeological Assessment*

12.6.46 Ngau Tam Mei (North) Area is identified to have high archaeological potential during the prehistoric period in the approved NOL Main Line and STLMC DN EIA reports (**Section 12.4.38** refers). It is located in the north of the assessment area of this Project.

12.6.47 Ngau Tam Mei (North) Area is situated on the southern hillslope of Ngau Tam Shan within the NTM river valley. The dominant geological composition of the Ngau Tam Mei (North) Area consists of colluvial deposits (*Qpd*), with some presence of alluvial deposits (*Qa* and *Qpa*) in the southern portion. In addition, the Area is positioned at higher levels, its elevation ranging between approximately +14 mPD and +28 mPD. The slopes in this Area are relatively steep, with gradients spanning from approximately 2° to 15°. Considering these geological and geographical features are conductive to past human settlements during the prehistoric period, Ngau Tam Mei (North) Area is determined to hold high archaeological potential. In similar environmental setting like Tsat Sing Kong SAI (SAI47), which is also sited on a river terrace in front of hillslope, archaeological features and artefacts dated to the Bronze Age were found[62].

12.6.48 The portion falls within the Project Site with high archaeological potential is hereinafter referred to as Ngau Tam Mei (North) ASA.

12.6.49 The current assessment has reviewed in detail and reassessed the results of previous assessments with old aerial photos and topographic maps. Some areas have been further reviewed in earlier sections and confirmed to have no archaeological potential (Plates 11 and 12 in **Appendix 12.2** refer) [63] [64]. For the areas where no clear modern disturbance captured in old aerial photos and topographic maps, particularly Tam Mei Barracks which were constructed in early 1950s, this Report will adopt the more conservative assessment result from the STLMC DN EIA report.

Chuk Yuen Tsuen Area

- *Previous Archaeological Assessment*

12.6.50 Chuk Yuen Tsuen Area was identified to have high archaeological potential in the NOL Main Line EIA report (**Section 12.4.36** refers). It is located in the west of the assessment area, which situated to the left of the Project Site across San Tin Highway. Chuk Yuen Tsuen is a traditional village established in 17th century, human activities and settlement during historical period was normal. Therefore, Chuk Yuen Tsuen holds high archaeological potential from historical period.

12.6.51 Chuk Yuen Tsuen Area falls outside the Project Site but within the 300 m assessment area.

[62] 香港考古學會（2009）。《元朗七星崗考古調查工作報告》。香港：古物古蹟辦事處檔案。
[63] Lands Department. (1974). *1:4800, 2400 ft., 09416 [aerial photo]*. Lands Department.
[64] Lands Department. (1975). *1:1 200 75-SW-C (Ed 1975-07) [topographic map]*. Lands Department.

12.6.52 The current assessment has reviewed the result of previous assessments and conducted a detailed review within the Project Site. Some areas have been further reviewed in earlier paragraphs and assessed to have no archaeological potential or low archaeological potential instead. Other than those areas previously discussed, this AIA agrees the archaeological potential assessment result of previous study over the remaining for Chuk Yuen Tsuen Area.

12.6.53 Furthermore, the assessment area of this AIA exceeded beyond that in the NOL Main Line EIA report to the west. However, other than the village area of Chuk Yuen Tsuen that has high archaeological potential in the historical period, the remaining areas lies on estuarine deposits and marine deposits (i.e. *QHH* and *Qam* shown in **Figure 12.2**), and thus contains no archaeological potential due to past submersion.

Siu Hum Tsuen Area

- *Previous Archaeological Assessment*

12.6.54 Siu Hum Tsuen Area was identified to have high archaeological during the prehistoric period in the approved NOL Main Line and STLMC DN EIA reports (**Section 12.4.38** refers). It is located in the north of the assessment area across the hills of Ngau Tam Shan. It has a river terrace landscape on gentle hill slopes that would have favoured past settlement. Geologically, it is located on a higher elevation between +2 mPD and +48 mPD (mean = +25 mPD) and on steep slopes between 1° and 18° (mean = 8°) approximately. It is further away from the coast, over 1,000 m to 2,000 m away. The geology of the area is mainly comprised of alluvial and colluvial deposits (*Qa*, *Qpa* and *Qpd*), features typical of river terrace. Its landscape is similar to other SAIs with known archaeological potential of prehistoric period, such as Sheung Shui Wa Shan SAI (SAI71).

12.6.55 Siu Hum Tsuen Area falls outside the Project Site but within the 300 m assessment area.

12.6.56 The current assessment has reviewed the result of previous assessments and conducted a detailed review within the Project Site. Some areas have been further reviewed in earlier paragraphs and assessed to have no archaeological potential instead. Other than those areas previously discussed, this AIA agrees the archaeological potential assessment results of previous studies over the remaining for Siu Hum Tsuen Area.

Long Ha Area

- *Previous Archaeological Assessment*

12.6.57 Long Ha Area was identified to have high archaeological potential in the NOL Main Line EIA report (**Section 12.4.36** refers). It is located in the south-east of the assessment area. The area has an elevation of +3 to +67 mPD (mean = +14 mPD) and gradient of 0° to 31° (mean = 5°) approximately. It is sitting between the mouth of two river valleys, Ngau Tam Mei and Kam Tin. It is located immediately in front of the hillslope of Kai Kung Leng facing westward. This Area is very close to the former coastline, within 500 m. Such coastal settings are similar to the westward facing SAIs with known archaeological potential of prehistoric period in Tuen Mun and Yuen Long, e.g., Ngau Hom Shek SAI (SAI37).

12.6.58 Long Ha Area falls outside the Project Site but within the 300 m assessment area.

12.6.59 The current assessment has reviewed the result of previous assessments and conducted a detailed review within the Project Site. Some areas have been further

reviewed in earlier paragraphs and assessed to have no archaeological potential. Other than those areas previously discussed, this AIA agrees the archaeological potential assessment result of previous study over the remaining for Long Ha Area.

Identification and Evaluation of Impacts

12.6.60 Potential impacts to archaeological heritage are identified in relation to the Project where construction works involving soil disturbance would occur during the construction phase. The latest site conditions are provided in **Appendix 12.3**.

Construction Phase - Direct Impact

12.6.61 As assessed in the previous archaeological surveys, Ngau Tam Mei SAI is confirmed to have high archaeological potential during the prehistoric and historical periods. Ngau Tam Mei SAI is partially located within the Project Site where land use of Government (UniTown) (G.11) and Road D1 are planned in the RODP (**Figure 12.8** refers). Direct impact to archaeological deposits (if any) would be anticipated for this area.

12.6.62 Both Ngau Tam Mei and Ngau Tam Mei (North) ASAs are assessed to have high archaeological potential and located within the Project Site. According to the RODP, these ASAs encroach on land uses of Private and Public Residential (R.1, R.2, R.3, R.4 and RSc.1), Education (E.2), Government, Institution or Community (G.3, G.4, G.7, G.8 and G.13), Government (UniTown) (G.6, G.10 and G.12), Open Space (O.2, O.3, O.4 and O.7), Green Belt (GB.2), Ngau Tam Mei Road, Road L1, L2, L3 and D1. Direct impact to the archaeological deposits (if any) would be anticipated during construction phase.

12.6.63 The upper stream low-lying agricultural fields and hilly landscape within the Project Site are considered to have moderate-low archaeological potential. These moderate-low archaeological potential areas encroach on land uses of Government, Institution or Community (G.7, G.8. and G.9), Government (UniTown) (G.11 and G.12), Road L3 and Road D1. Direct impact to the archaeological deposits (if any) would be anticipated during construction phase.

Construction Phase - Acceptable Impact

12.6.64 For the low archaeological potential areas identified within the Project Site, direct impact to the potential archaeological remains (if any) is anticipated. However, the impact on archaeology is considered minimal and acceptable.

Construction Phase - No Impact

12.6.65 The areas that have unfavourable landscape features for human settlements have been confirmed to lack archaeological potential in previous archaeological surveys and assessments, or have been disturbed heavily due to modern development (such as large-scale infrastructure works, property developments and fishponds), are therefore considered to have no archaeological potential. No impact to archaeological remains within these areas is anticipated.

12.6.66 With reference to **Section 2.10**, *NOL Main Line*[65] and *STLMC DN*[66] would be concurrently implemented along with this Project. Some of their works sites and works areas will encroach into the Project Site (**Figure 2.5** refers). Provided that

[65] MTR Corporation Limited. (2023). *Environmental Impact Assessment Study for Northern Link (Agreement No. C1603). Section 13 Cultural Heritage Impact Assessment.* Retrieved from https://www.epd.gov.hk/eia/files/applications/en/pp_346/eia_1992/progress/action_1365/eia_3012023/HTML/EIA%20Report/S13%20CHIA.htm.
[66] *Ibid*.

the archaeological impacts, if any, would have been identified and mitigated by the concurrent projects before handover to this Project, no archaeological impacts are anticipated at these concerned areas during the construction phase.

12.6.67 For areas outside the Project Site, as no construction works will be conducted at these areas, no impact to archaeological remains (if any) at these areas is anticipated.

Operational Phase

12.6.68 No impact to archaeological remains is anticipated during operational phase as any archaeological impacts would have been mitigated prior or during the construction phase.

Mitigation Measures

Pre-Construction Phase

*Archaeological Excavation at Ngau Tam Mei SAI (**Figures 12.11** and **12.12** refer)*

12.6.69 Archaeological excavation is necessary to be conducted at the Ngau Tam Mei SAI within the Project Site to obtain adequate archaeological information of the area, and retrieve the archaeological data, if any, before commencement of works involving soil disturbance at the respective area. It should be subject to future land resumption status and discussion with AMO in later stages.

12.6.70 The scope, methodology and programme of the archaeological excavation should be agreed with AMO, and the excavation should be conducted by an archaeologist who should have obtained a *Licence to Excavate and Search for Antiquities* from the Secretary for Development prior to the commencement of the fieldworks under A&MO. Should archaeological deposits discovered in the archaeological fieldworks, mitigation measures should be proposed and agreed with AMO.

*Archaeological Survey-cum-Excavation at Archaeologically Sensitive Areas (**Figures 12.11** and **12.12** refer)*

12.6.71 This CHIA has conducted a comprehensive review based on all available information at this stage, in order to safeguard any archaeological resources that might be existed within the Project Site. Considering the potential direct impact to both Ngau Tam Mei ASA and Ngau Tam Mei (North) ASA with high archaeological potential has been identified, archaeological survey-cum-excavation should be conducted. The aim is to obtain adequate archaeological information of these areas for verifying their archaeological potential, and retrieve the archaeological data, if any, before commencement of works involving soil disturbance at the respective areas. The programme of survey-cum-excavation should be subject to future land resumption status and discussion with AMO in later stages.

12.6.72 In principle, archaeological survey should be conducted with an aim to locate the precise horizontal extent and nature of the archaeological deposits. Should key archaeological findings occurred, excavation works should be applied to retrieve archaeological data completely before the commencement of works involving soil disturbance at the respective areas.

12.6.73 The scope, methodology and programme of the survey-cum-excavation should be agreed with AMO, and the survey-cum-excavation should be conducted by an archaeologist who should have obtained a *Licence to Excavate and Search for Antiquities* from the Secretary for Development prior to the commencement of the

fieldworks. Should archaeological deposits discovered in the archaeological fieldworks, mitigation measures should be proposed and agreed with AMO.

*Archaeological Survey at Moderate-Low Archaeological Potential Areas (***Figures 12.11** and **12.12** *refer)*

12.6.74 Considering the potential direct impact to both the low-lying agricultural fields and hilly landscape within the Project Site where possess moderate-low archaeological potential, archaeological survey should be conducted. The aim is to obtain adequate archaeological information of these areas for verifying their archaeological potential, and retrieve the archaeological data, if any, before commencement of works involving soil disturbance at the respective areas. The survey programme should be subject to future land resumption status and discussion with AMO in later stages.

12.6.75 The scope, methodology and programme of the archaeological survey should be agreed with AMO, and the survey should be conducted by an archaeologist who should have obtained a *Licence to Excavate and Search for Antiquities* from the Secretary for Development prior to the commencement of the survey under A&MO. Should archaeological deposits discovered in the archaeological survey, mitigation measures should be proposed and agreed with AMO.

Construction Phase

12.6.76 The low archaeological potential areas identified in the AIAs under the NOL Main Line and STLMC DN EIA studies within the Project Site would be subject to acceptable impact. As a precautionary measure and pursuant to the A&MO (Cap. 53), the project proponent is required to inform the AMO immediately in case of discovery of antiquities or supposed antiquities in the course of works, so that appropriate mitigation measures, if needed, can be timely formulated and implemented in agreement with and to the satisfaction of AMO.

12.6.77 For the areas that have been disturbed heavily due to modern development and have no archaeological potential, no impact to archaeological heritage by the proposed works is anticipated. As a precautionary measure and pursuant to the A&MO (Cap. 53), the project proponent is required to inform the AMO immediately in case of discovery of antiquities or supposed antiquities in the course of works, so that appropriate mitigation measures, if needed, can be timely formulated and implemented in agreement with and to the satisfaction of AMO.

12.6.78 As construction works will only be conducted within the Project Site, there is no anticipated impact on other archaeological potential areas located outside Project site. Therefore, no mitigation measure is required.

Operational Phase

12.6.79 As no impact is anticipated during operational phase, no mitigation measure is required.

12.6.80 A summary on the potential impact and the proposed mitigation measures for archaeological heritage is presented in **Table 12.3**.

Environmental Monitoring and Audit

12.6.81 The implementation of the recommended mitigation measures mentioned in **Sections 12.6.69** to **12.6.78** should be audited as part of the EM&A programme. Details of the EM&A requirements are provided in a stand-alone EM&A Manual.

Conclusion

12.6.82 Based on the desktop review and the findings of previous archaeological surveys, there are high archaeological potential areas located within the Project Site, namely Ngau Tam Mei SAI, Ngau Tam Mei ASA and Ngau Tam Mei (North) ASA. Direct impact on archaeological heritage is anticipated during the construction phase.

12.6.83 Archaeological excavation is recommended to be conducted at the Project Site within Ngau Tam Mei SAI in order to obtain adequate archaeological information of the area, and retrieve the archaeological data, if any, before commencement of works involving soil disturbance at the respective area, subject to future land resumption status and discussion with AMO in later stages.

12.6.84 This CHIA has conducted a comprehensive review based on all available information at this stage, in order to safeguard any archaeological resources that might have existed within the Project Site. Considering the potential direct impact to both Ngau Tam Mei ASA and Ngau Tam Mei (North) ASA where have high archaeological potential, archaeological survey-cum-excavation is recommended to be conducted at these areas. The aim is to obtain adequate archaeological information of these areas for verifying their archaeological potential, and retrieve the archaeological data, if any, before commencement of works involving soil disturbance at the respective areas. It should be subject to future land resumption status and discussion with AMO in later stages.

12.6.85 Considering the potential direct impact to both the low-lying agricultural fields and hilly landscape within the Project Site where possess moderate-low archaeological potential, archaeological survey should be conducted to obtain adequate archaeological information of these areas for verifying their archaeological potential, and retrieve the archaeological data, if any, before commencement of works involving soil disturbance at the respective areas. It should be subject to future land resumption status and discussion with AMO in later stages.

12.6.86 The low archaeological potential areas identified within the archaeological assessments in the NOL Main Line and STLMC DN EIA reports within the Project Site is considered as having acceptable impact, while there would be no impact on archaeological heritage by the Project for the areas that have been disturbed heavily due to modern development and have no archaeological potential. As a precautionary measure and pursuant to the A&MO (Cap. 53), the project proponent is required to inform the AMO immediately in case of discovery of antiquities or supposed antiquities in the course of works, so that appropriate mitigation measures, if needed, can be timely formulated and implemented in agreement with and to the satisfaction of AMO.

12.6.87 As construction works will only be conducted within the Project Site, there is no anticipated impact on other archaeological potential areas located outside the Project Site. Therefore, no mitigation measures are required.

Table 12.3 Summary of Potential Impact and the Proposed Mitigation Measures on Archaeological Heritage in the Vicinity of the Project Site

Area of Concern	Impact Assessment	Impact Level	Mitigation Measures
Ngau Tam Mei SAI	As assessed in previous archaeological investigations, Ngau Tam Mei SAI is confirmed to have **high** archaeological potential during the prehistoric and historical period. Land uses of Government (UniTown) (G.11) and Road D1 as planned in the RODP encroached into the Ngau Tam Mei SAI. Direct impact to archaeological deposits (if any) would be anticipated for this area.	Direct Impact	Archaeological excavation is necessary to be conducted at the Project Site within Ngau Tam Mei SAI to obtain adequate archaeological information of the area, and retrieve the archaeological data, if any, before commencement of works involving soil disturbance at the respective area. It should be subject to future land resumption status and discussion with AMO in later stages.
Ngau Tam Mei ASA and Ngau Tam Mei (North) ASA	Ngau Tam Mei and Ngau Tam Mei (North) ASA are assessed to have **high** archaeological potential located within the Project Site. According to the RODP, land uses of Private and Public Residential (R.1, R.2, R.3, R.4 and RSc.1), Education (E.2), Government, Institution or Community (G.3, G.4, G.7, G.8 and G.13), Government (UniTown) (G.6, G.10 and G.12), Open Space (O.2, O.3, O.4 and O.7), Green Belt (GB.2), Ngau Tam Mei Road, Road L1, L2, L3 and D1 are located within both Ngau Tam Mei Archaeological Potential Area and Ngau Tam Mei (North) Archaeological Potential Area. Direct impact to the archaeological deposits (if any) would be anticipated during the construction phase.	Direct Impact	This CHIA has conducted a comprehensive review based on all available information at this stage, in order to safeguard any archaeological resources that might have existed within the Project Site. Considering the potential direct impact to both Ngau Tam Mei ASA and Ngau Tam Mei (North) ASA where have high archaeological potential, archaeological survey-cum-excavation should be conducted. The aim is to obtain adequate archaeological information of these areas for verifying its archaeological potential, and retrieve the archaeological data, if any, before commencement of works involving soil disturbance at the respective area. It should be subject to future land resumption status and discussion with AMO in later stages. In principle, archaeological survey-cum-excavation should be conducted with an aim to locate the precise horizontal extent and nature of the archaeological deposits. Should key archaeological findings occurred, excavation works should be applied to retrieve archaeological data completely before commencement of works involving soil disturbance at the respective area.

Area of Concern	Impact Assessment	Impact Level	Mitigation Measures
Moderate-Low Archaeological Potential Area	The low-lying agricultural fields and hilly landscape within the Project Site are considered to have **moderate-low** archaeological potential. These moderate-low archaeological potential areas encroach on land uses of Government, Institution or Community (G.7, G.8 and G.9), Government (UniTown) (G.11 and G.12), Road L3 and Road D1. Direct impact to the archaeological deposits (if any) would be anticipated during construction phase.	Direct Impact	Considering the potential direct impact to both the low-lying agricultural fields and hilly landscape where have moderate-low archaeological potential, archaeological survey should be conducted. The aim is to obtain adequate archaeological information of these areas for verifying its archaeological potential, and retrieve the archaeological data, if any, before commencement of works involving soil disturbance at the respective area. It should be subject to future land resumption status and discussion with AMO in later stages.
Low Archaeological Potential Area	For the **low** archaeological potential areas identified in the NOL Main Line and STLMC DN EIA Reports within Project Site, direct impact to the potential archaeological remains (if any) is anticipated. However, the impact on archaeology is considered minimal and acceptable.	Acceptable Impact	As a precautionary measure and pursuant to the A&MO (Cap. 53), the project proponent is required to inform the AMO immediately in case of discovery of antiquities or supposed antiquities in the course of works, so that appropriate mitigation measures, if needed, can be timely formulated and implemented in agreement with and to the satisfaction of AMO.
No Archaeological Potential Area	Desktop study has revealed that the Project Site has undergone extensive modern disturbance, primarily driven by infrastructure projects. Additionally, the construction of property developments, cut slopes and fishponds have contributed to the disturbance. These modern construction activities had involved significant soil disturbance, resulting in the disturbance of the natural soil (and hence any potential archaeological deposits). As a result, the developed or disturbed areas within the Project Site are considered to have **no** archaeological potential.	No Impact	As a precautionary measure and pursuant to the A&MO (Cap. 53), the project proponent is required to inform the AMO immediately in case of discovery of antiquities or supposed antiquities in the course of works, so that appropriate mitigation measures, if needed, can be timely formulated and implemented in agreement with and to the satisfaction of AMO.

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